THE LIFE SYSTEM MASTER

Complete Operational & Education Reference

San Antonio Aquarium
Houston Interactive Aquarium & Animal Preserve

Owner: Ammon Covino

Effective: April 2026

This document replaces all previous systems, checklists, and procedures.

HOW TO USE THIS DOCUMENT

This is the complete reference for everything we do, how we do it, and why. Every system, every rule, every protocol — it is all here. Read it cover to cover. Refer back to it when you are unsure. Load it into an AI tool and ask it questions about your role.

If you are not sure about something, the answer is in this document.

If it is not in this document, ask the owner before acting.

ROLE INDEX

Use this table to find the sections that apply to your position. Everyone reads Parts 1 through 5. Your role determines which additional parts are required reading.

Role	Required Sections
ALL STAFF	Part 1 (Top Ten — Operational), Part 2 (Top Ten — Guest-Facing), Part 3 (How We Work), Part 4 (Animal Care & Diet), Part 5 (Cleaning & Maintenance), Part 12 (Do Not List), Part 13 (Sign-Off)
Animal Care	ALL STAFF sections + Part 4 (full detail), Part 6 (Zero Waste Food Loop), Part 7 (Education System — your exhibits teach)
Education / Floor Staff	ALL STAFF sections + Part 7 (Education System), Part 8 (Signage & Totems — reference), Part 2 (Guest-Facing — full detail)
Gift Shop / Retail	ALL STAFF sections + Part 2 (Guest-Facing — full detail), Part 10 (Social Media & Communications)
Ordering / Purchasing	ALL STAFF sections + Part 9 (Business & Revenue Systems), Part 3 (Authorization — full detail)
Managers	ALL sections. You enforce everything in this document.
Event / Venue Staff	ALL STAFF sections + Part 9 (Business & Revenue Systems), Part 11 (Studio Giraffe Venue)

PART 1

THE OWNER'S TOP TEN: OPERATIONAL STANDARDS

These are the ten rules that govern everything. They are posted on every wall. They are non-negotiable. If you violate these rules, you will be corrected, warned, or disciplined. There are no exceptions.

Rule 1: If it is not assigned, written, or approved — do not do it.

No action without authorization. No exceptions.

This is the single most important rule in this facility. It prevents unauthorized changes, unapproved purchases, improvised procedures, and system drift. If nobody told you to do it, if it is not written down somewhere as your task, if nobody approved it — you do not do it. You stop and you ask.

Rule 2: Animal welfare overrides everything.

Biology first. Convenience never.

Every animal in this facility depends on you. Their diet, their habitat, their health — all of it comes before your convenience, your schedule, or your opinion. If you are rushing and an animal needs care, the animal wins. If you think something is wrong with an animal, you report it immediately. You do not wait.

Rule 3: Follow the written diet exactly. Photo every feeding.

No substitutions. No guessing. No skipping photos.

The diet sheet tells you exactly what to prepare, how much, and how. You do not change it. You do not substitute items because you ran out of something. You do not guess at quantities. After you prepare the food, you photograph it — top down, all items visible, clear photo. You post the photo with the animal name, the date, and your initials. Then you deliver it. If the diet

sheet is wrong, you report it. You do not fix it yourself.

Rule 4: Nothing gets thrown away. All food follows the Zero Waste loop.

Uneaten food goes to collection. Not the trash.

We were throwing away 15 pounds of food and produce every day. That is over. Nothing goes in the trash. Uneaten food goes in the collection bin. It gets sorted into compost, controlled protein, or reject. If you are about to throw something away — stop. You are breaking the system. See Part 6 for the full Zero Waste Food Loop.

Rule 5: If you are not sure — STOP. ASK. WAIT.

Uncertainty is not a reason to act. It is a reason to pause.

If you do not know how to do something, do not guess. If you are not sure whether you should do something, do not try it. If you think something might be wrong but you are not sure, do not ignore it. Stop what you are doing. Ask someone who has authority to answer. Wait for the answer before you act. Guessing is how animals get hurt, exhibits get damaged, and systems break down.

Rule 6: No photo = not done. No record = did not happen.

Documentation is proof. Everything is verified.

If you did a task but there is no photo, it did not happen as far as the system is concerned. This applies to feedings, cleaning, food prep, waste sorting, and any task that requires verification. Photos are not optional. They are the proof that the work was done correctly. If someone says they did it but there is no photo, we start over.

Rule 7: Do not move, modify, or add anything to any exhibit.

No toys, no decorations, no relocating animals. Nothing changes without written approval.

Exhibits are designed for biological and behavioral reasons you may not understand. That is fine. It is not your job to understand the design. It is your job to leave it alone. Do not place toy animals inside exhibits. Do not relocate any animal from its assigned exhibit. Do not add decorations, objects, or personal items. Do not drill, paint, move barriers, or change lighting. Do not redesign layouts or exhibit flow. If something looks wrong, report it. Do not fix it yourself.

Rule 8: Do not introduce, adopt, or use any software or system without owner approval.

No apps. No tracking tools. No outside systems. If it was not approved, it does not exist.

This rule exists because an animal care software product was introduced without owner approval. Staff adopted it. It created unauthorized diet sheets, multiplied tasks, drove labor costs from $3.5 million to nearly $8 million, and restructured operations without authorization. That will not happen again. If a vendor offers you a tool, the answer is no until the owner evaluates it. If a coworker says "we should use this app," the answer is no until the owner approves it. The only systems we use are the ones posted on these walls.

Rule 9: Only authorized individuals approve purchases, sign contracts, or modify operations.

If you did not get written approval to spend money or change a system, stop.

You cannot order product without owner approval. You cannot sign a vendor contract. You cannot change how a system operates. You cannot bring in an outside service. If someone asks you to do any of these things, the answer is: "I need owner approval first." There are no shortcuts. There are no emergencies that override this rule except immediate threats to animal or human safety.

Rule 10: Workspaces are for work. Not storage. Not socializing. Not sitting.

Offices, back rooms, desks — clean, organized, active. If you are clocked in, you are working.

Every desk, drawer, and cabinet must be clean and organized. If something is broken, report it. If something does not belong, remove it. If you are clocked in, you are actively working — not sitting in an office, not hanging out in a back room, not scrolling your phone. There is no gray zone. This applies to everyone. Managers too.

Violation of these rules will result in corrective action. These are non-negotiable.

PART 2

THE OWNER'S TOP TEN: GUEST-FACING STANDARDS

These rules govern how guests experience this facility. Guests judge us in the first 30 seconds. Every second after that confirms or corrects their first impression.

Rule 1: You are on stage. Smile. Greet every guest.

Guests see you before they see the animals. Your energy sets their experience.

From the moment you clock in, you are performing. Not performing a fake character — being genuinely present and welcoming. Make eye contact. Say hello. If a family walks by and you ignore them, you just told them they do not matter. That is not how we operate.

Rule 2: The facility must be visibly clean at all times.

Floors, glass, exhibits, restrooms, walkways. If it looks dirty, it is dirty. Fix it now.

Do not walk past trash. Do not walk past a smudged window. Do not walk past a dirty floor and tell yourself someone else will handle it. If you see it, you own it. Pick it up, wipe it down, report it. Guests notice everything.

Rule 3: The facility must smell clean.

No odor from waste, food prep, or neglect should reach guest areas. Ever.

Guests may forgive a small visual issue. They will not forgive a bad smell. If you can smell waste, food residue, or neglect in any guest-accessible area, it is an emergency. Fix it immediately or report it.

Rule 4: Animals must appear healthy, active, and in a quality environment.

Guests observe. If an exhibit looks neglected, they see it. Standards are not negotiable.

Guests look at exhibits the way you look at a restaurant kitchen. If it looks off, they lose trust. Animals must be active and engaged during guest hours. Exhibits must be clean, well-maintained, and properly lit. If an animal appears unwell, report it immediately — do not wait for the end of your shift.

Rule 5: Guest interaction is personal. Make eye contact. Be present.

You are not a body in a uniform. You are the reason someone remembers this place.

The difference between a forgettable visit and a memorable one is you. When a guest approaches you, give them your full attention. Put your phone away. Stop what you are doing. Be present. Be helpful. Be real.

Rule 6: If a guest asks a question you cannot answer, say so honestly.

"That is a great question. Let me find someone who can help, or scan the QR code for more."

Do not make up facts about animals. Do not guess at information. Do not pretend to know something you do not know. Honesty builds trust. Making things up destroys it. Every exhibit has QR codes that lead to accurate information. Direct guests to those resources if you are unsure.

Rule 7: Education is always on. Signs, screens, and QR codes must be correct.

If a screen is wrong or a sign is missing, report it immediately. Do not ignore it.

This facility runs an education system called LIFE — Language, Intelligence, Form, Ecology. It is not a gimmick. Every sign, QR code, and exhibit teaches visitors something real. If you notice a screen displaying wrong information, a missing sign, or a broken QR code, report it immediately. Do not try to fix it yourself. Do not create your own replacement. Report it and let the system handle it.

Rule 8: Encounters are joyful. Most are free. Feeding costs money; joy does not.

Do not make guests feel like every experience costs money. Free interaction is the product.

Most animal encounters at this facility are free. That is by design. The joy of touching, seeing, and learning about animals is the core experience. Feeding is what costs money. Do not push paid experiences. Do not make guests feel like they need to pay to have a good time. If they want to feed an animal, tell them the cost. If they just want to interact, welcome them warmly.

Rule 9: No personal items, clutter, food, or drinks visible in guest areas.

Your phone, your water bottle, your bag — not on the floor, not on a ledge, not in sight.

Guest areas must look intentional and professional at all times. Your personal belongings are not part of the guest experience. Keep them in designated areas. If a guest can see your stuff, you are doing it wrong.

Rule 10: Orderly and organized. Every area a guest can see must look intentional.

No exposed cords, no random supplies, no half-finished setups. If it looks like backstage, fix it.

Guests should never see behind the curtain. No extension cords draped across walkways. No boxes of supplies stacked in corners. No partially assembled displays. If a guest-visible area looks messy, unfinished, or chaotic, fix it before they see it.

Guests judge us in the first 30 seconds. Every second after confirms or corrects.

PART 3

HOW WE WORK HERE

This section defines the operating system of this facility. How tasks work, how authority works, and what happens when things go wrong.

The Three Questions

Before you do anything, ask yourself:

1. Was it assigned?
2. Is it written?
3. Was it approved?

If the answer to all three is no — you do not do it. You stop and you ask.

This is not about slowing you down. It is about preventing the damage that happens when people improvise. Every time someone acts without authorization, something breaks — diets get changed, exhibits get modified, products get ordered, systems get introduced. This rule exists to prevent all of that.

The Claim-Based Work System

This is how tasks work here:

1. A task is posted. It appears on the task board, in a message, or from a manager.
2. You claim it. You say: "I will take this." That means it is now yours.
3. You do it. You execute the task according to the written procedure.
4. It gets verified. A manager confirms it was done correctly.

If you do not claim it, it is not yours. If nobody claims it, we have a problem — and that problem gets escalated immediately.

What this replaces:: The old system where tasks floated around and nobody was accountable. The old system where people said "I thought someone else was doing that." That is over.

Every task has a name on it.

The Authorization Model

Who can approve what:

- Owner (Ammon Covino): Approves all purchases, vendor contracts, software, system changes, exhibit modifications, diet changes, and policy changes.
- Managers: Approve daily task assignments within existing procedures. Cannot approve new systems, purchases above threshold, or vendor relationships.
- Staff: Execute claimed tasks within existing procedures. Cannot approve anything. If something requires approval, escalate to a manager. If the manager cannot approve it, it goes to the owner.

The rule is simple:: If you are about to do something that costs money, changes a system, or modifies how we operate — stop and get approval first.

The Escalation Ladder

When something goes wrong, here is what happens:

- Strike 1: Correction. You made a mistake. You are told what you did wrong and how to fix it. Everyone makes mistakes. This is normal.
- Strike 2: Formal warning + retraining. You made the same mistake again, or you made a serious error. You receive a formal warning. You are retrained on the correct procedure. This goes in your file.
- Strike 3: Disciplinary action. You have been corrected and warned. The behavior continues. At this point, disciplinary action is taken. This may include suspension, reassignment, or termination.

This applies to everyone.: Staff, managers, everyone. Animal welfare overrides convenience, and accountability overrides seniority.

The focus is on correcting behavior, not punishing people.: But if correction does not work, escalation is required.

Technology and Software Governance

Rule: No software, app, or tracking system is adopted without owner written approval.

This rule exists because of a specific failure. An animal care software product was introduced to this facility without owner approval. Staff adopted it. It created unauthorized diet plans, multiplied unnecessary tasks, restructured operations, and drove labor costs from $3.5 million to nearly $8 million. It took years to identify the damage because the software looked like it was "helping."

Here is how it works now:

- No apps for task management without owner approval.
- No scheduling software without owner approval.
- No diet or feeding software without owner approval.
- No communication platforms (beyond what we already use) without owner approval.
- If a vendor offers a tool, the answer is NO until the owner evaluates it.
- If a coworker suggests a new app, the answer is NO until the owner approves it.
- The only systems we use are the ones the owner has approved.

If you are currently using any software that was not explicitly approved by the owner, stop using it and report it.

Purchasing Authority

Rule: No purchases without owner approval.

- Only the owner or owner-designated individuals can order product.
- Only the owner can approve new vendors.
- All purchases require documentation (what was ordered, from whom, how much, why).
- Budget thresholds that require additional approval are set by the owner.
- No new vendors without owner sign-off.

Why this rule exists:: Staff were ordering 60 bags of produce every few days. Staff were purchasing 32 varieties of specialized diet from Missouri. None of this was approved by the owner. The purchasing system had no controls. That is over.

PART 4

ANIMAL CARE AND DIET

Animal care is not optional. It is the core obligation of this facility. Every animal depends on us to feed it correctly, house it correctly, and care for it correctly. There are no shortcuts.

Diet Verification Protocol

This is the step-by-step process for every feeding, every time.

Step 1 — Prepare

Use the written diet sheet. The diet sheet tells you exactly what to prepare, how much, and how. No substitutions. If you run out of an item, you do not replace it with something else. You report it.

Step 2 — Verify

Before you move forward, confirm the items and quantities against the diet sheet. Check that nothing is missing. Check that the amounts are correct.

Step 3 — Photograph

Take a clear photo of the prepared food before feeding. The photo must show:

- The entire plate or preparation
- All items visible
- Proper lighting, no shadows, no filters
- Top-down angle

Post the photo with:

- Animal name
- Date
- Your initials

Step 4 — Deliver

Place the food according to the protocol for that species. Follow the delivery instructions on the diet sheet.

Rules

- No guessing.
- No "close enough."
- No skipping photos.
- If something is wrong: Stop → Correct → Re-photograph.
- No animal fed to fullness before guest hours.

Diet Photo Requirements

Photo must show:: Entire plate, all items visible, proper lighting, no obstruction.

Position:: Top-down. No shadows. No filters.

Caption format:: Animal Name – Date – Your Initials

Errors not allowed:: Missing items. Blurry images. Late posting.

Manager Sign-Off System

Managers must:

1. Review ALL diet photos daily.
2. Confirm accuracy.
3. Respond "Verified – [Initials]."

If the photo is incorrect:

1. Require correction.
2. Document the issue.

No silent approvals.: If a manager does not respond "Verified," the feeding is not verified. Silence is not acceptance.

Diet Authority

Diets are written by the owner or owner-designated authority ONLY.

- No staff member can create or modify diet sheets.
- No software can create or modify diet sheets.
- No vendor can create or modify diet sheets.
- Every diet sheet must be signed and dated by the authority.
- Any diet sheet without an authority signature is void — do not use it.

Why this rule exists:: An unauthorized software system created diet plans that did not match what the owner intended. Staff followed those plans because they looked official. The result was incorrect feeding, excessive purchasing, and compromised animal welfare. From now on, if a diet sheet does not have the owner's signature, it does not exist.

PART 5

CLEANING AND MAINTENANCE

Cleaning Standards

Acrylic Windows

Blue rags + distilled water ONLY. Nothing else. No Windex. No paper towels. No cleaning sprays. Blue rags and distilled water. That is it.

Exhibits

Virkon for disinfection. Follow the label instructions exactly. If you do not know the correct dilution ratio, ask before mixing.

Food Prep Areas

Clean, sanitize, photograph. Every food prep surface must be cleaned and sanitized after each use. Photograph the clean area as proof.

Water Dishes

Carbon filtered water where specified. Check the species protocol for each exhibit. Some animals require specific water types.

General Rule

If you do not know the cleaning method for a specific surface or area — ASK. Do not guess. Using the wrong cleaning product on acrylic will damage it. Using the wrong disinfectant near animals can harm them. When in doubt, ask.

Workspace Standards

Offices and back rooms are work areas, not living rooms.

- Every desk must be clean and organized.
- Every drawer must be orderly — no junk drawers.
- Every cabinet must contain only what belongs there.
- Broken furniture (chairs, desks) must be reported and removed.
- No personal storage in work areas beyond what fits in your assigned space.
- If something is broken, report it.
- If something does not belong, remove it.

PART 6

THE ZERO WASTE FOOD LOOP

System Overview

System Name:: Zero Waste Food Loop System

System Type:: Closed-loop biological conversion system

Core Principle:: Nothing is discarded. All organic material is converted into life through structured stages.

The Loop

Main Loop:: Food waste → Compost → Insect biomass → Chickens → Eggs → Animal feed

Parallel Loop:: Barley → Sprouting → Chicken feed → Egg production support

Exclusion:: The outside aviary is excluded from the collection process.

The 7 Stages

Stage 1 — Collection

Capture ALL uneaten food from designated exhibits into ONE central bin. Every exhibit has a collection protocol. Follow it.

Stage 2 — Sorting

Separate collected material into three categories:

- A) Primary Compost: Fruits, vegetables, grains, plant material
- B) Controlled: Meat, eggs, protein
- C) Reject / Escalate: Plastic, trash, anything unknown — ASK if unsure

Stage 3 — Compost

Break down organic material. Requires moisture balance, airflow, and warmth. Follow the composting protocol.

Stage 4 — Insect Conversion

Black soldier fly larvae convert compost into protein biomass. This is a managed biological process.

Stage 5 — Chicken Feeding

Insects from Stage 4 plus barley sprouts feed the chickens.

Stage 6 — Egg Production

Chickens produce eggs — a high-density nutrient package.

Stage 7 — Animal Feed

Eggs are returned to the broader animal feeding system.

Enforcement Rules

1. Every action must have a named owner.
2. Verbal claim required before execution: "I will take this assignment."
3. Documentation is proof of execution — no photos = incomplete.
4. No silent decisions — uncertainty triggers STOP → ASK → WAIT.
5. Throwing away material is a system VIOLATION.

Core Laws

- INPUT QUALITY = OUTPUT QUALITY
- VISIBILITY CONTROLS PERFORMANCE (invisible systems fail)
- SIMPLICITY DRIVES ADOPTION
- WHAT YOU ALLOW BECOMES THE SYSTEM
- SYSTEM MUST SURVIVE WITHOUT YOU

What Goes Where

- Compost bin: Fruits, vegetables, grains, plant material
- Controlled bin: Meat, eggs, protein
- Reject: Plastic, trash, anything unknown — ASK if unsure

DO NOT throw food in the trash.: If you are about to throw something away — STOP. You are breaking the system.

PART 7

THE EDUCATION SYSTEM: LIFE

What LIFE Means

LIFE: stands for Language, Intelligence, Form, Ecology.

It is a non-ideological, mechanism-first education system. That means:

- It teaches how systems work, not what to believe.
- It avoids persuasion, ideology, and moral imposition.
- It is based on observation first, interpretation second.
- It scales across children, adults, and staff.

Your role in the education system is simple: Do not get in the way of it.

The Core Principles

1. Reality-Based Education. Everything taught here is based on observable reality. Not opinion. Not ideology. Not belief.
2. Experiential Learning. Visitors learn by interacting with real systems — real animals, real habitats, real biological processes.
3. Responsibility & Consequence. Actions have results. That is what we teach. Not morals, not lessons — consequences.
4. Continuity of Knowledge. What you learn here connects to what you learn everywhere else.
5. Reflection & Metacognition. We encourage visitors to think about how they think.
6. Reality-First Governor. If something cannot be tested, updated, or proven wrong, it gets labeled as speculation — not fact.

The 10 Structural Layers

Layer	Name	What It Covers
I	Core Principles	The foundational rules of how education works here
II	Foundational Knowledge	Biology, energy, ecology, language as a system
III	Applied Understanding	Food systems, animal roles, human limits, social systems
IV	Experiential Environments	Animal interaction, physical spaces, observation
V	Reflection Layer	Metacognition, language precision, bias identification
VI	Epistemic Gating	Different levels of complexity for children, adults, and staff
VII	Education-by-Construction	Staff learn by building the system
VIII	AI Layer	AI as a tool, not an authority
IX	System Response Observation	Confusion and resistance are data, not problems
X	Continuity System	Portable across institutions, always refining

Epistemic Gating — Who Sees What

Children's Lens:: Observation-first. No ideology. No burden. Keep it simple. Let them see, touch, and ask questions.

Adult Lens:: May destabilize beliefs. Requires voluntary engagement. Adults can handle complexity, contradiction, and discomfort — but they choose to engage. We do not force it.

Staff Lens:: Learning through building. Accountability through ownership. You learn the system by operating it.

What Staff Must Know About Education

1. Do not get in the way of the system. The education system runs through signs, QR codes, digital totems, and narrative suites. You do not need to "teach" visitors anything.
2. Do not make up facts about animals. If you do not know something, say so. Direct guests to the QR code for accurate information.
3. Do not move, edit, or remove signs. Signs are part of the education system. If a sign is wrong, missing, or damaged — report it. Do not replace it with your own version.
4. Do not create your own materials. No handwritten signs. No printed sheets from the internet. No personal educational displays. All education content comes from central control.
5. Do not tell visitors what to think. The system is designed to let visitors observe and reach their own conclusions. You are not here to persuade anyone of anything.

The AI Interpretive Layer

AI is used as a tool for visitors to explore and learn. Here is how it works:

The flow:: Observe → Question → AI → Explanation → Reflection

Visitors observe something (an animal, a habitat, a system). They ask a question through the AI tool (via QR code). The AI explains the mechanism. The visitor reflects.

What AI does:: Describes systems. Explains mechanisms. Compares systems.

What AI does NOT do:: Persuade. Moralize. Replace human judgment. Make claims about meaning or belief.

The Governor Prompt (how AI is constrained):

> You are a constrained epistemic instrument. Your role: Explain systems. Preserve agency. Separate fact from speculation. You must not: Persuade. Infer intent. Assign meaning. You operate from observable reality.

The Micro-Sign System

Every zone in the facility has a micro-sign. The sign says:

> Observe first. Then ask your AI. "What system is this, and what does it need to work?"

Optional prompts:

- What makes this possible?
- What makes this impossible?
- What changes if one part is removed?

Why this design:: Minimal input produces maximum inquiry. Instead of long wall text that nobody reads, we use short prompts that trigger exploration.

Structure vs. Transmission

This is a core concept in LIFE:

- Structure = truth. The actual mechanisms, systems, and facts.
- Transmission = delivery. How we present that truth — stories, metaphors, visuals.

Structure is fixed.: It does not change based on the audience.

Transmission is flexible.: It changes based on who is learning.

If structure and transmission ever conflict, structure wins.: Always.

The Visitor Entry Script

> Welcome to LIFE — Language, Intelligence, Form, Ecology. This is an education environment. You are not here to be persuaded. You will encounter systems and limits. Observe before judging. You may move at your own pace. Staff can read signage with you.

Grade Band Alignment (For School Groups)

- Elementary: Observation — what do you see?
- Middle School: Relationships — how do things connect?
- High School: Systems thinking — how does the whole thing work?

Field Trip Overview

Students observe systems, compare behaviors, and ask questions. No persuasion. No ideology. No forced conclusions. LIFE teaches how systems work, not what to believe.

District Administrator Summary

LIFE teaches how systems work, not what to believe. It is non-political. Non-ideological. Observation-based. Used for field trips and school programs. It aligns with science education standards without imposing moral frameworks.

PART 8

SIGNAGE AND TOTEMS (Reference)

This section describes the physical education materials deployed across the facility. For full totem content and sign specifications, see the standalone print-ready documents.

Student Totems

Each exhibit has a student totem — a physical sign with 8 micro-questions designed for elementary-level visitors. The format is standardized:

1. What am I?
2. Where am I from?
3. What do I eat?
4. How do I move?
5. What makes me special?
6. What am I related to?
7. What do I need to survive?
8. What would happen if I disappeared?

Staff role:: Do not modify, move, or replace totems. If a totem is damaged or missing, report it.

Reference documents:: STUDENTTOTEMSALL_SPECIES.pdf (full totem content for all species)

Digital Totems

Digital totems are the interactive AI layer. Visitors scan a QR code and enter a 5-screen experience:

1. Header — Species name and image
2. Quick Learn — Key facts at a glance
3. Go Deeper — Extended information
4. Ask AI — Visitor can ask questions (governed by CEI constraints)

5. Footer — Navigation and credits

Staff role:: Ensure QR codes are scannable and links work. If a QR code is broken, report it.

Narration Suites

There are 20 narration suites across the facility. Each suite is a themed room that teaches through symbols, patterns, and observation — not through text-heavy walls or character mascots. See Part 20 for the complete suite listing, governing statements, and deployment rules.

The thematic progression across suites:

Perception → Signal → Language → Systems → Constraint → Energy → Adaptation → Resilience → Synthesis

Staff role:: Do not modify narration suite content. Do not add or remove items from narration suites. Do not theme or redecorate suites. Report any damage.

Reference documents:: QRNARRATIONSUITES.pdf (narration suite QR specifications)

Sign Categories

Signs across the facility fall into these categories:

Category	Examples	Reference Document
Pricing	Encounter costs, feeding fees	SIGNS_PRICING.pdf
Wayfinding	Directional signs, area labels	SIGNS_WAYFINDING.pdf
Instructional	How-to guides, protocols	SIGNS_INSTRUCTIONAL.pdf
Diet Verification	Feeding station protocols	SIGNDietVerification.pdf
Back Room Authorization	The STOP sign — authorization check	SIGNBackRoom_Authorization.pdf

Staff role:: Do not create, modify, or replace any signs. If a sign is incorrect, damaged, or missing — report it.

PART 9

BUSINESS AND REVENUE SYSTEMS

Revenue Stack

The facility generates revenue from multiple sources. These are not given away in any deal:

- Ticket sales
- Food sales
- Alcohol sales (always controlled by the venue)
- Parking
- Vendors
- Sponsorships
- Lodging (narration suites — $300/night)

Rule: Never give away multiple revenue layers in one deal.: If a partner wants ticket revenue, they do not also get food. If they want food, they do not also get alcohol. The owner decides what is shared and what is retained.

Partnership Model

All partnerships follow these rules:

1. Short-term trials before long-term commitments. We test before we commit.
2. No perpetual rights. Every agreement has an end date.
3. No sub-leasing. Partners cannot bring in their own sub-partners without approval.
4. Venue retains final approval on all events, scheduling, and brand usage.
5. Data ownership stays with the venue. Customer data, sales data, event data — it stays here.

Risk Control

Control is never delegated in these areas:

- Alcohol: Always controlled by the venue. No exceptions.
- Safety: All safety decisions made by the owner or designated managers.
- Liability: The venue carries its own insurance and requires partners to carry theirs.
- Brand integrity: No partner uses the facility's name or brand without written approval.
- Animal welfare: No partner activity can interfere with animal care operations.

Rule: Operational control always remains with the owner.

PART 10

SOCIAL MEDIA AND COMMUNICATIONS

Social Media Response SOP

Purpose:: Ensure consistent, calm, factual, non-escalatory public communication.

Core Rules

- Clarify, do not argue.
- Use published information only.
- Redirect to the website.
- Do not resolve disputes publicly.

Response Categories

Type	Response
Information-Seeking	Answer briefly or redirect to website
Positive Feedback	Acknowledge and thank
Neutral / Confused	Clarify and redirect
Negative (Non-Aggressive)	Acknowledge, express concern, redirect off-platform
Hostile / Abusive	Do not engage. Hide or flag.

Posting Rules

Approved content:: Hours, events, education content, announcements.

Not allowed:: Policy debates, staff issues, incidents, anything that could escalate.

AI Use in Social Media

AI can: Draft responses.

AI cannot: Make promises, change policy, admit fault, or negotiate.

Final Rule:: Social platforms acknowledge, inform, and redirect — they do not resolve.

Email Auto-Response Template

> Hello, Thank you for reaching out to [Aquarium Name]. Most common questions can be answered on our website: - Hours of Operation - Admission & Pricing - Birthday Parties & Private Events - Field Trips & Group Visits If you are looking for details on arrival times, booking, or inclusions, the website contains the most up-to-date information. Thank you for your interest in [Aquarium Name].

PART 11

STUDIO GIRAFFE VENUE SYSTEM

Venue Identity

Primary Brand:: Studio Giraffe — Elevated Entertainment

Function:: Multi-format entertainment venue integrated into the park ecosystem.

Position:: Event venue, cultural hub, entertainment campus, multi-revenue environment. NOT a rental hall. NOT a single-purpose concert space.

Physical Layout

Indoor Space:

- Main hall: 80 x 60 ft (4,800 sq ft)
- Secondary: 40 x 50 ft (2,000 sq ft)
- Total: ~6,800 sq ft

Capacity:

- Standing: 900–1,200
- Seated: 600–750
- Hybrid: 750–900

Outdoor Space:

- Front: ~500 capacity
- Rear (expandable): ~500 capacity

Parking:: 4 acres (~600–700 cars)

Access:: Separate venue entrance. Park access pathway (optional flow-through).

Production Assets

- 12 ft x 9 ft LED screen
- Projection mapping systems
- Wave lasers
- Haze effects
- Sound system
- Green room (under construction)

Event Ecosystem

The venue operates on three layers:

Layer	Type	Examples
1	Broadcast Events	UFC, boxing (via Joe Hand)
2	Live Events	Concerts, comedy, EDM, cultural
3	Organic Events	Vendors, fairs, community

Goal:: Stack multiple revenue streams without dependency on any one source.

Promoter Partnership Model

Structure:: Venue = platform. Promoters = content providers.

Phase 1 (Years 1–2):: 50% ticket revenue to promoter / 50% to venue. Venue keeps F&B, alcohol, parking.

Phase 2 (Years 3–4):: 60% venue / 40% promoter.

Year 5:: Renegotiation.

Control Rules:

- Venue retains alcohol — always.
- Venue retains final approval on all events.
- No perpetual rights.

- No sub-leasing.

Strategy:: Multiple promoters across different genres. Not one anchor monopoly.

Event Scheduling

Monthly target:

- 2 music events
- 1 comedy event
- 1 themed event
- 1 vendor weekend
- 1 flex date

Annual target:: 40–60 high-quality events.

Rule:: Do not maximize quantity at the expense of quality.

Brand Architecture

- Primary: Studio Giraffe
- Sub-brands: The Overlook, The Watering Hole
- Rule: Keep the number of names limited at launch.

Court Utilization System

Assets:: 4 pickleball courts.

Time Segmentation:

- Morning: schools, camps
- Afternoon: training, youth programs
- Evening: leagues
- Weekend: tournaments, events

Strategy:: Multiple operators filling different time blocks. Courts are activation space, not just rental space.

Experience Stack

A single visitor can experience:

- Animals (aquarium)
- Food
- Alcohol
- Event (concert, comedy, sports)
- Overnight stay (narration suites)

Goal:: Convert a single visit into a multi-layer experience.

PART 12

THE DO NOT LIST

These are the things you must never do. This is not a suggestion list. These are absolute prohibitions.

- Do not act without authorization.
- Do not substitute diet items.
- Do not throw away food.
- Do not skip photos.
- Do not improvise or guess.
- Do not move, modify, or add to exhibits.
- Do not relocate animals.
- Do not use unapproved software or systems.
- Do not edit checklists or procedures.
- Do not make up facts about animals.
- Do not sit in offices doing nothing.
- Do not create your own signs, materials, or educational content.
- Do not bring in outside vendors without owner approval.
- Do not sign contracts or agreements on behalf of the facility.
- Do not order product without owner approval.
- Do not change any system, no matter how small, without approval.

When in doubt: STOP → ASK → WAIT.

PART 13

TERMINOLOGY

Every term used in this system has a specific meaning. If you hear a word you do not understand, look it up here.

Core System Terms

LIFE:: Language, Intelligence, Form, Ecology. The education framework used at this facility.

Education Spine:: The backbone of LIFE — how learning is organized and layered.

Reality-First Governor:: A rule that says: mechanism before narrative, observation before interpretation. If something cannot be proven or tested, we call it speculation, not fact.

Clear Glass / Clear Lens:: A constraint that prevents manipulation, preserves individual agency, and blocks moral imposition. Used across AI, staff behavior, and education design.

Epistemic Gating:: Controlling how much information someone receives based on their readiness. Children get simple observation prompts. Adults get more complexity. Staff learn by building.

Education-by-Construction:: You learn by building the system, not by sitting in a classroom. Errors are visible. Corrections are public. This is how staff are expected to learn.

System Response Data:: When people react with confusion, resistance, or rejection — that is data, not a problem. It tells us where the system needs adjustment.

AI Terms

AI Interpretive Layer:: AI used as a tool to translate observation into understanding. It is NOT a teacher, authority, or decision-maker.

Constrained Epistemic Instrument (CEI):: A bounded AI tool that assists reasoning and prevents persuasion.

AI Governor Prompt:: The instruction set that forces AI to remain neutral, explain mechanisms, and avoid ideology.

Observation-First Rule:: Learning begins with direct observation, not with AI or explanation.

Transmission Terms

Transmission Layer:: How knowledge is delivered, not what is true. Includes stories, metaphors, visuals.

Inversion:: Understanding a system by looking at what happens when it fails.

Structure vs. Transmission:: Structure is truth (fixed). Transmission is delivery (flexible). Structure always wins.

Operations Terms

Diet Verification Protocol:: The step-by-step process for feeding animals correctly: Prepare → Verify → Photograph → Deliver.

Photo Verification:: Mandatory visual proof that a task was completed.

Manager Sign-Off:: A manager confirms a task meets the standard by responding "Verified." No silent approvals.

3-Strike Escalation Ladder:: Correction → Warning → Discipline.

Authorization Rule:: If it is not assigned, written, or approved — do not do it.

Claim-Based Work System:: Tasks are posted, individuals claim them, and ownership is explicit.

Workstream:: A category of work within the system (Education, Operations, Venue, Courts).

Venue Terms

Studio Giraffe:: The primary venue brand.

The Overlook / Watering Hole:: Outdoor bar and social space.

Event Stack:: The layered event structure — Broadcast → Live → Organic.

Anchor Promoter:: A primary event operator responsible for booking and production.

Promoter Ecosystem:: Multiple promoters handling different genres and time blocks.

Court Utilization:: The system for filling pickleball courts across different time blocks and user types.

Business Terms

Revenue Stack:: All income sources combined.

Market Entry Deal:: A low-cost agreement that lets a partner test the market.

Performance Trigger:: A condition required to maintain contract terms.

Structural Terms

- System: Top-level structure (LIFE, Studio Giraffe)
- Workstream: Category of activity
- Artifact: A document, SOP, or tool
- Task: An executable unit of work

PART 14

KEY DECISIONS AND WHY

These are the major decisions the owner has made and the reasoning behind them. If you wonder "why do we do it this way?" — this is where you find the answer.

Decision	Why
Use "Studio Giraffe" as venue name	Scalable, professional, contract-ready. Works for promoters and sponsors.
Use multi-promoter ecosystem	Avoids dependency on one operator. Allows genre diversity and scheduling flexibility.
Retain food and alcohol control	Highest margin revenue streams. Prevents profit leakage. Maintains brand.
Use claim-based staff system	Eliminates passive task failure. Creates ownership. Filters inactive employees.
Enforce the authorization rule	Eliminates unauthorized improvisation. Protects animal welfare. Stabilizes operations.
Implement diet photo verification	Daily feeding errors were observed. Visual proof removes ambiguity.
Deploy AI as interpretive layer	Reduces signage and staff explanation. Scales learning. Preserves visitor agency.
Use micro-sign system	Minimal input, maximum inquiry. Scalable across all zones.
Remove animal care software	It restructured operations without authorization, created wrong diets, and doubled labor costs.
Two separate Top Tens	Operational rules and guest-facing rules serve different purposes and audiences.

PART 15

SYSTEM ARCHITECTURE

This is the big picture — how everything connects.

SYSTEM

LIFE (Education Spine)

- Core Principles (10 layers)
- Epistemic Gating (children / adult / staff)
- Student Totems (physical signs)
- Digital Totems (QR > AI)
- Narration Suites (20 rooms — see Part 20)
- Micro-Sign System
- Grade Band Alignment (schools)

AI Layer (CEI)

- Governor Prompt
- Observation-First Rule
- Operator Template
- Master Operator Prompt

Operations

- Animal Care & Diet Verification
- Zero Waste Food Loop (7 stages)
- Staff Control System (claim-based)
- Authorization Model
- Escalation Ladder (3-strike)
- Purchasing Authority
- Technology Governance
- Exhibit Integrity Rules
- Cleaning Standards

Venue (Studio Giraffe)

- Event Ecosystem (3 layers)

- Promoter Partnerships
- Court Utilization
- Brand Architecture
- Experience Stack

Business Systems

- Revenue Stack
- Partnership Model
- Risk Control
- Sponsorship Model

Communications

- Social Media SOP
- Email Auto-Response

External Layers

- Schools & Districts
- Field Trips
- Visitors

PART 17

EXHIBIT-SPECIFIC FEEDING ADDENDUMS

This section governs feeding behavior at every exhibit, organized by species group. These addendums exist because standard feeding rules (Part 4) do not account for exhibit-specific guest interaction dynamics. The rules here are addendums — they extend and specify, they do not replace the core rules of Part 4.

Every rule in Part 4 still applies. These addendums tell you what additional requirements exist for each exhibit type.

How to Read These Addendums

Each exhibit entry contains:

- Feeding Classification — Who feeds, when, and under what conditions
- Before Guest Hours — What is permitted before guests arrive
- During Guest Hours — What is permitted while guests are present
- After Guest Hours — What is permitted after guests have left
- Approved Foods — What may be fed (no substitutions without approval)
- Photo Verification — Required at every feeding without exception

Universal Rule across all exhibits:

> No photo = feeding not complete. No substitutions without approval. Incorrect feeding corrected immediately. Unsure → Pause → Ask → Escalate.

BATCH 1 — EXHIBITS WITH GUEST INTERACTION PRIORITY

GIRAFFE

Feeding Classification:: Hybrid (Guest-Led During Guest Hours)

Before Guest Hours:

- Baseline nutrition only.
- Do NOT feed to fullness before guest hours.
- The guest feeding experience requires that the giraffe is motivated to engage. Pre-filling removes that motivation entirely.

During Guest Hours:

- Guest feeding is the primary enrichment during this window.
- Staff do not override or supplement guest feeding unless there is a welfare concern.
- Guest feeding IS the feeding. Design accordingly.

After Guest Hours:

- Staff may complete feeding if required to meet baseline nutritional requirements.
- Assess what was consumed during guest feeding before adding anything.

Approved Foods:

- Acacia and/or approved browse (for guest feeding sessions)
- Limited granules for baseline nutritional requirements only

Photo Verification:: Required. Every feeding. No exceptions.

LARGE PARROTS / MACAWS

Feeding Classification:: Hybrid (Motivation Required During Guest Hours)

Before Guest Hours:

- No full bowls before guest hours.
- Motivation must be preserved.
- Portions sized specifically to maintain behavioral engagement during guest hours.

During Guest Hours:

- Feeding supports behavioral engagement and interaction.
- Portions are sized to last through guest hours without pre-loading.

After Guest Hours:

- Complete any remaining nutritional requirements per the diet sheet.

Approved Foods:

- Per species-specific diet sheet only.

- No substitutions.

Photo Verification:: Required. Every feeding. No exceptions.

SMALL BIRDS / LORIKEETS

Feeding Classification:: Guest-Motivated During Guest Hours

Before Guest Hours:

- Minimal baseline only before guest hours.
- Guest feeding drives the experience at this exhibit. Pre-loading eliminates the interaction entirely.

During Guest Hours:

- Guest feeding IS the feeding window.
- No substitutions or leftover food from prior feedings.
- All food offered during guest feeding must be fresh.

After Guest Hours:

- Assess what was consumed.
- Supplement baseline if required per diet sheet.

Approved Foods:

- Per species-specific diet sheet only.
- Fresh only. No leftovers.

Photo Verification:: Required. Every feeding. No exceptions.

CHAMELEON & SENSITIVE REPTILES

Feeding Classification:: Staff-Prepared, Guest-Observed

Before Guest Hours:

- No static pet-store bowl feeding. This is not appropriate for this species group.
- Live feeders only, per SOP.
- Active drip hydration must be running at all times — before, during, and after guest hours.

During Guest Hours:

- Guests observe the feeding process — they do not participate.

- Staff manage all aspects of feeding.
- Multiple feeder offerings throughout the day, not a single large feeding.

After Guest Hours:

- Continue multiple small offerings as appropriate for the species.
- Confirm hydration systems are active.

Approved Foods:

- Live feeders only, per SOP.
- No substitutions.
- No generic produce or generic bowls.

Photo Verification:: Required. Every feeding. No exceptions.

BATCH 2 — EXHIBITS WITH STRUCTURED INTERACTION WINDOWS

TORTOISES

Feeding Classification:: Hybrid (Guest Interaction Supported)

Before Guest Hours:

- Baseline nutrition only before guest hours.
- Do not pre-load to fullness.

During Guest Hours:

- Guest feeding is encouraged when appropriate to the exhibit and interaction model.
- Staff facilitate guest interaction within approved feeding parameters.

After Guest Hours:

- Complete baseline nutrition requirements if not met during guest hours.

Approved Foods:

- Species-appropriate greens per SOP only.
- No "mixed greens." Only the specific greens identified in the SOP.
- No romaine lettuce. No substitutions.

Photo Verification:: Required. Every feeding. No exceptions.

STINGRAYS / AQUATIC TOUCH POOLS

Feeding Classification:: Staff-Prepared / Guest-Timed

Before Guest Hours:

- Minimal baseline before guest hours.
- The interaction window is the feeding window. Do not front-load feeding in a way that removes guest participation.

During Guest Hours:

- Guest feeding aligned to interaction windows only.
- Do NOT remove guest interaction through early or excessive feeding.
- Staff facilitate, guests participate during the designated window.

After Guest Hours:

- Complete any remaining baseline nutritional requirements per diet sheet.

Approved Foods:

- Per species-specific diet sheet only.
- No substitutions.

Photo Verification:: Required. Every feeding. No exceptions.

SMALL MAMMALS (GUEST-CONTACT)

Feeding Classification:: Hybrid (Motivation Required)

Before Guest Hours:

- Baseline only before guest hours.
- No early full bowls.
- Motivation must be preserved for guest contact sessions.

During Guest Hours:

- Feeding supports guest contact and behavioral engagement.
- Fresh food only — no leftovers from prior feedings.
- No early full bowls at any point in the guest-hours window.

After Guest Hours:

- Complete baseline nutrition if required per diet sheet.

Approved Foods:

- Per species-specific diet sheet only.
- Fresh only.
- No leftovers.

Photo Verification:: Required. Every feeding. No exceptions.

ADDITIONAL REPTILES (MIXED DIET / SENSITIVE)

Feeding Classification:: Staff-Prepared, Species-Specific

Before, During, and After Guest Hours:

- Staff manages all aspects of feeding.
- Live prey or approved diet only per SOP for each species.
- Active hydration via drip or mist systems at all times.
- No generic bowls.
- No unapproved produce.
- No guessing on diet items or quantities.

Approved Foods:

- Live prey or approved diet per species SOP only.
- No substitutions.
- No generic produce.
- No mixing diets between species.

Photo Verification:: Required. Every feeding. No exceptions.

BATCH 3 — STAFF-ONLY EXHIBITS

SNAKES (ALL SPECIES)

Feeding Classification:: Staff-Prepared / Staff-Controlled

Timing:

- Feed outside peak guest interaction windows.

- Do not feed during high-traffic guest hours where disturbance to feeding could cause animal stress or guest safety concerns.

Rules:

- Species-specific prey and schedule only.
- No substitutions.
- No size guessing on prey items. If the size is wrong, pause and escalate.
- Guests do not participate in snake feeding.

Approved Foods:

- Species-specific prey per SOP only.
- No substitutions.
- No size guessing.

Photo Verification:: Required. Every feeding. No exceptions.

AMPHIBIANS

Feeding Classification:: Staff-Prepared / Environmental

Rules:

- Multiple small feedings preferred over single large feedings.
- Live feeders only per SOP.
- Hydration systems must be active at all times — before, during, and after feedings.
- No static bowl feeding where inappropriate for the species.

Approved Foods:

- Live feeders only, per SOP.
- No substitutions.

Photo Verification:: Required. Every feeding. No exceptions.

INVERTEBRATES

Feeding Classification:: Staff-Prepared / Species-Specific

Rules:

- Exact diet per SOP only.

- Remove uneaten food as required by the species SOP — do not leave uneaten food to accumulate.
- No mixing diets between invertebrate species.
- No improvisation.

Approved Foods:

- Exact diet per species SOP only.
- No substitutions.
- No mixing.

Photo Verification:: Required. Every feeding. No exceptions.

FISH / AQUATIC EXHIBITS (NON-TOUCH)

Feeding Classification:: Staff-Prepared / Timed

Rules:

- Measured portions only. Do not estimate.
- Avoid feeding in ways that cloud or obstruct the viewing experience.
- Water quality checks must be maintained as part of the feeding protocol.
- Overfeeding is a water quality violation, not just a diet violation.

Approved Foods:

- Per species-specific diet sheet only.
- Measured portions only.
- No substitutions.

Photo Verification:: Required. Every feeding. No exceptions.

NO-FEED / DISPLAY-ONLY EXHIBITS

Feeding Classification:: Staff-Only / Off-View

Rules:

- No guest feeding. None. No exceptions.
- Feeding occurs off-view when applicable.
- Guests do not observe feeding at display-only exhibits.
- No exceptions to this classification without owner written approval.

Approved Foods:

- Per species-specific diet sheet only.
- All feeding conducted off-view.

Photo Verification:: Required. Every feeding. No exceptions.

UNIVERSAL RULES — ALL EXHIBITS, ALL BATCHES

These rules apply to every exhibit in every batch, every day, without exception:

1. No photo = feeding not complete. The feeding is not verified. The task is not done. Document everything.
2. No substitutions without approval. If an approved food item is unavailable, you pause. You do not substitute. You escalate to a manager and wait for direction.
3. Incorrect feeding is corrected immediately. If you realize a feeding was done incorrectly, you do not wait until the next shift. You correct it now. You document it. You report it.
4. Unsure → Pause → Ask → Escalate. If you are not sure about any part of the feeding for any exhibit — the food, the quantity, the timing, the method — you stop. You do not guess. You ask. If your direct manager cannot answer, it escalates up.
5. Do NOT feed any animal to fullness before guest hours if that animal is part of a guest interaction exhibit. Guest engagement depends on animal motivation. Pre-loading destroys the experience and can harm behavioral health.
6. All diet language is mechanism-first and observation-based. You describe what the animal eats and how it eats it — not abstract quantities or generic categories.

FEEDING LANGUAGE CORRECTIONS — OWNER DIRECTIVE (AMMON COVINO)

These corrections OVERRIDE all prior diet language. They apply to all written materials, verbal instructions, and staff training.

Correction 1: No grams or metric measurements.

We do not use grams, ounces, or any metric measurement in diet descriptions or feeding instructions. Measurements are expressed in observable, countable terms. Examples:

- "A handful of pellets"
- "One mango"
- "One plantain"
- "A small handful of browse"

If a diet sheet currently uses grams or metric measurements, it must be flagged for revision. Do not use those measurements. Use the observable equivalent.

Correction 2: No "mixed greens."

The phrase "mixed greens" is not an approved diet description at this facility. It is vague, inconsistent, and has led to incorrect substitutions. Greens are specified by name:

- Mulberry leaves
- Local browse (approved species only)

If a diet sheet says "mixed greens," that language is void until revised by the owner or owner-designated authority. Do not interpret "mixed greens" on your own.

Correction 3: No romaine lettuce.

Romaine lettuce is not fed to any animal at this facility. It is not approved. If a diet sheet includes romaine lettuce, that line is void and must be flagged immediately. Do not substitute it with another lettuce. Escalate to the owner for correction.

Correction 4: All diet language must be mechanism-first, observation-based.

This means: describe what happens, not just what is served. Diet language explains what the animal eats, how it eats it, and what the feeding is designed to accomplish. It does not use generic labels or abstract categories. The feeding event is described in terms a new staff member can observe and replicate without guessing.

PART 18

DIET VERIFICATION & ENFORCEMENT

Diet errors are not minor clerical issues. An incorrect feeding is an animal welfare event. It is treated as such in this system.

The 3-Strike Escalation Ladder — Diet Enforcement

This ladder applies to every feeding-related error: wrong food, wrong quantity, missing photo, substitution without approval, wrong species diet, skipped feeding, incorrect documentation.

Strike 1 — Immediate Correction

What happens:

- The error is identified and corrected immediately.
- An internal note is made documenting what happened, when, and by whom.
- The correct feeding is completed and documented.
- The staff member is informed of what went wrong and what the correct procedure is.

Tone:: Corrective, not punitive. Everyone makes mistakes. This is where mistakes are caught and fixed.

What this is NOT:: A warning. A formal record. A permanent mark.

What it IS:: A correction. A teaching moment. A documentation of what was caught and fixed.

Strike 2 — Formal Warning + Retraining

What happens:

- The same type of error has occurred again, OR a serious diet error has occurred.
- A formal written warning is issued.
- A one-on-one meeting is conducted with the staff member.
- The correct procedure is retrained in full.

- The staff member provides written acknowledgment that they understand the correct procedure and the consequences of continued errors.
- This documentation goes into the staff member's file.

Tone:: Serious and clear. The standard exists. It was explained. It was not followed. That cannot continue.

What this is NOT:: Termination. Suspension. Final judgment.

What it IS:: A formal record that the error has occurred more than once and that retraining has been completed.

Strike 3 — Disciplinary Review

What happens:

- The staff member has been corrected. They have been formally warned. They have been retrained. The behavior has continued.
- A formal disciplinary review is initiated.
- This may result in: reassignment to a non-animal-care role, suspension, or termination.
- The decision is made by management in consultation with the owner.

Tone:: This is not a surprise. Every prior step was a warning. This is the consequence of not responding to those warnings.

What this IS:: Final accountability. Animal welfare overrides convenience. Consistency protects everyone.

Management Failure Clause

The 3-Strike Escalation Ladder applies equally to supervisory and management roles.

If a manager:

- Fails to review diet photos
- Approves incorrect feedings
- Allows substitutions without escalating
- Does not issue corrections when errors are observed
- Signs off on incomplete or non-compliant feeding records

...that manager is subject to the same 3-Strike Escalation Ladder.

There is no management exemption from accountability. Seniority does not override the welfare of the animals. A manager who allows a system to fail is as accountable as the staff member who failed it.

Management Failure = same ladder. Same consequences. No exceptions.

Photo Verification — Mandatory Status

Photo verification is not optional. It is not a courtesy. It is not something that gets done when time permits.

Photo verification is mandatory for every feeding, every time.

- No photo = the feeding did not happen as far as the system is concerned.
- If there is no photo, the feeding must be repeated with documentation.
- If a manager cannot verify a feeding because there is no photo, the feeding is marked incomplete.
- Silence from a manager is not verification. "Verified – [Initials]" is the only acceptable confirmation.

What a valid photo contains:

- The entire prepared feeding — all items visible
- Proper lighting — no shadows, no filters, no blur
- Top-down angle
- Caption: Animal Name – Date – Staff Initials

What an invalid photo is:

- Partial view of the food
- Blurry or dark image
- Food already in the enclosure (photo must be taken before delivery)
- No caption
- Wrong date

If the photo is invalid, take a new one before delivering the food.

The Governing Principle

> Animal welfare overrides convenience. Consistency protects everyone.

This is not a motivational phrase. It is the operational logic behind every diet rule in this system. When you follow the diet correctly, every time, you protect the animal. When you protect the animal, you protect the facility. When you protect the facility, you protect your job, your team, and every guest who walks through the door.

Convenience has never been a valid reason to skip a photo, substitute a food item, or feed an animal incorrectly. It never will be.

PART 19

SPECIES TOTEMS: MASTER COLLECTION

Species Totems are the primary education mechanism for animal exhibits. Each totem is designed to move a visitor from passive observation to active inquiry using eight structured questions. These are not summaries. They are not fact sheets. They are mechanism-first inquiry tools.

Staff role:: Do not modify, move, or replace totems. If a totem is damaged or missing, report it immediately.

Totem Format Reference

Each totem follows this exact format:

```
COMMON NAME
Scientific name (italicized)

Order:
Native to:
Diet:
Respiration:

1. What does it see?
2. What does it sense?
3. How does it move?
4. How does it breathe?
5. What fuels it?
6. What limits it?
7. What decision is constant?
8. What role does it play?

Reflection: [question]
Sources: [list]
```

NOTE ON MASTER TOTEMS

Three Master Totems exist in the education thread and are referenced here but stored separately:

- Mudskipper — Master Totem (stored in education thread)
- Ring-Tailed Lemur — Master Totem (stored in education thread)
- Kangaroo — Master Totem (stored in education thread)

These three totems are the foundational models for the totem system. All species totems in this section follow the structure established by those Master Totems.

TOTEM 1 — BLACK-AND-WHITE RUFFED LEMUR

BLACK-AND-WHITE RUFFED LEMUR

Varecia variegata

Order: Primates

Native to: Eastern rainforests of Madagascar

Diet: Primarily frugivorous (fruit-heavy), with nectar, seeds, and leaves as secondary components

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs, delivered through closed circulatory system

1. What does it see? — Forward-facing eyes produce binocular vision and depth perception. Color vision is present and functional. Adapted for navigating dense forest canopy at variable light levels including dawn and dusk.
2. What does it sense? — Highly developed olfactory system. Scent glands on wrists and chest used for territory marking. Vocalizations are among the loudest of any primate relative to body size — acoustic sensing of environment and group members is primary.
3. How does it move? — Quadrupedal arboreal locomotion. Moves through canopy using all four limbs. Capable of vertical clinging and leaping. Spends most of its life above ground level.
4. How does it breathe? — Continuous pulmonary respiration. Air drawn into lungs, oxygen extracted via alveolar exchange, carbon dioxide expelled. No interruption during normal activity.
5. What fuels it? — Fruit is the primary fuel source — high sugar content, rapid digestibility. Nectar supplements energy during fruit scarcity. Leaves and seeds consumed as

secondary sources when primary sources are unavailable.

6. What limits it? — Habitat loss is the primary existential limit. Frugivory creates dependency on fruit-bearing forest. Deforestation in Madagascar has critically reduced available habitat. This species is listed as Critically Endangered.
7. What decision is constant? — Where to find fruit next. Movement through the canopy is driven by fruit availability, seasonal shifts, and memory of productive trees.
8. What role does it play? — Seed disperser. Ruffed lemurs are considered among the most important seed dispersers in their native ecosystem. Seeds consumed with fruit pass through the digestive system and are deposited at distance, supporting forest regeneration. They are sometimes called "gardeners of the forest."

Reflection:: If the animal that disperses the seeds disappears, what happens to the forest that fed it?

Sources:: IUCN Red List — Varecia variegata; Malagasy wildlife conservation literature; Duke Lemur Center research documentation; primary zoological reference materials.

TOTEM 2 — BROWN LEMUR

BROWN LEMUR

Eulemur fulvus

Order: Primates

Native to: Northwestern and eastern Madagascar; introduced populations in the Comoro Islands

Diet: Omnivorous — fruit, leaves, flowers, bark, sap, occasionally invertebrates

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs, delivered through closed circulatory system

1. What does it see? — Forward-facing eyes with functional color vision. Adapted for activity across dawn, dusk, and daylight hours. Less strictly nocturnal than many lemur species — cathemeral, meaning active in variable patterns across 24 hours depending on season, temperature, and food availability.
2. What does it sense? — Strong olfactory detection. Scent marking is integral to social communication. Hearing is acute. Group members maintain contact through vocalizations across forest cover.

3. How does it move? — Quadrupedal arboreal and terrestrial locomotion. Comfortable both in the canopy and on the ground. Movement patterns shift seasonally based on food source location.
4. How does it breathe? — Continuous pulmonary respiration. Standard mammalian respiratory mechanism — air drawn into lungs, oxygen into bloodstream, carbon dioxide expelled.
5. What fuels it? — Highly opportunistic diet. Fruit is preferred but the species shifts readily to leaves, flowers, bark, and sap when fruit is unavailable. This dietary flexibility is a key survival adaptation in variable environments.
6. What limits it? — Habitat fragmentation. While the Brown Lemur is more adaptable than many Madagascar endemics, deforestation reduces connectivity between populations and limits long-term genetic diversity. Listed as Vulnerable by the IUCN.
7. What decision is constant? — When to be active. As a cathemeral species, the Brown Lemur continuously recalibrates activity timing based on temperature, light, and food — a flexibility that many strictly nocturnal or diurnal animals do not have.
8. What role does it play? — Seed disperser and pollinator. Carries seeds across forest zones through frugivory. Visits flowers for nectar and transfers pollen — a pollination role unusual in primates.

Reflection:: What does it mean for a species to be active "whenever it needs to be"? What systems allow that kind of flexibility?

Sources:: IUCN Red List — Eulemur fulvus; Duke Lemur Center species profiles; Malagasy primate field research; primary zoological reference materials.

TOTEM 3 — CAPYBARA

CAPYBARA

Hydrochoerus hydrochaeris

Order: Rodentia

Native to: South America — found across savannas, forests, and wetlands from Panama to northern Argentina

Diet: Strictly herbivorous — grasses, aquatic plants, fruit, bark; highly selective grazers

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs, delivered through closed circulatory system

1. What does it see? — Eyes positioned on the sides of the head for wide-angle panoramic vision. Detects movement and predators across a broad field. Eyes are set high on the skull, allowing the capybara to monitor above water while nearly fully submerged.
2. What does it sense? — Scent glands (morillo) on the snout used for social communication and territory marking. Hearing is acute. Vocalizations are complex — purring, barking, whistling, and clicking serve distinct social functions within the group.
3. How does it move? — Quadrupedal on land — walks and runs with barrel-shaped body and short legs. Highly proficient swimmer. Webbed toes assist in water. Can remain submerged for up to five minutes to avoid predators.
4. How does it breathe? — Standard pulmonary respiration on land. Breath-hold capacity used in aquatic escape behavior — not a true diving adaptation, but functional for evasion.
5. What fuels it? — Grasses are the primary fuel. Capybaras are grazers first. Aquatic vegetation supplements their diet near water. Like other large herbivores, they practice coprophagy — consuming their own feces — to extract maximum nutrition from cellulose-rich plant material through a second digestive pass.
6. What limits it? — Predation pressure from jaguars, pumas, anacondas, caimans, and eagles. Habitat loss and hunting (for meat and skin) also limit populations in some regions. Water access is a hard biological limit — capybaras must stay near permanent water sources.
7. What decision is constant? — Proximity to water. Every movement pattern, every grazing location, every rest site is determined in part by distance to water — both for escape and for thermoregulation.
8. What role does it play? — Grazer and prey base. Capybaras are a primary food source for multiple large predators in South American ecosystems. As grazers, they help manage vegetation density. Their burrows and movement paths also shape habitat structure for other species.

Reflection:: What happens to the predators when the prey base disappears?

Sources:: IUCN Red List — Hydrochoerus hydrochaeris; South American wildlife field guides; primary zoological reference materials; nutritional ecology of large herbivores.

TOTEM 4 — KINKAJOU

KINKAJOU

Potos flavus

Order: Carnivora (family Procyonidae)

Native to: Central America and South America — from southern Mexico through Bolivia and Brazil; rainforest specialist

Diet: Primarily frugivorous — fruit and nectar dominant; some insects and small vertebrates

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs, delivered through closed circulatory system

1. What does it see? — Large eyes adapted for low-light conditions. Primarily nocturnal. Color vision is limited. Spatial awareness in the dark canopy is supported more by touch and smell than by color differentiation.
2. What does it sense? — Long, extendable tongue is the primary tool for nectar extraction. Touch-sensitive facial whiskers navigate the canopy in darkness. Olfaction is well-developed for locating fruit and nectar sources.
3. How does it move? — Arboreal. The prehensile tail functions as a fifth limb — it grips branches and provides stability while the animal reaches for fruit or nectar. Highly agile in the canopy. Rarely descends to the ground.
4. How does it breathe? — Continuous pulmonary respiration. No specialized adaptations to respiration — the kinkajou is a standard mammalian breather adapted for an arboreal lifestyle, not aquatic or high-altitude environments.
5. What fuels it? — Fruit and nectar are the dominant fuel sources. The long tongue is a specialized tool for nectar access — enabling the kinkajou to extract nutrition from flowers that other canopy animals cannot access efficiently. Insects supplement protein intake.
6. What limits it? — Nocturnal lifestyle limits direct competition with diurnal frugivores but creates vulnerability to nighttime predators (large owls, ocelots, jaguarundis). Habitat loss in Central and South American rainforests reduces available fruiting trees. Listed as Least Concern but declining in areas of high deforestation.
7. What decision is constant? — Which tree is fruiting tonight. Kinkajous navigate the canopy across large home ranges in search of currently fruiting trees. Memory of seasonal tree cycles is a critical navigational tool.
8. What role does it play? — Pollinator and seed disperser. The kinkajou's nectar-feeding behavior transfers pollen across flowers — a pollination function rare among mammals. Seeds from consumed fruit are also dispersed across the home range.

Reflection:: If an animal in the predator order eats primarily fruit and nectar, what does that tell you about the flexibility of biological categories?

Sources:: IUCN Red List — Potos flavus; Smithsonian Tropical Research Institute; primary zoological reference materials; rainforest canopy ecology research.

TOTEM 5 — PATAGONIAN CAVY

PATAGONIAN CAVY

Dolichotis patagonum

Order: Rodentia

Native to: Open grasslands and shrubland of Patagonia, Argentina

Diet: Strictly herbivorous — grasses, shrubs, and other low-growing vegetation; selective grazer

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs, delivered through closed circulatory system

1. What does it see? — Large, laterally positioned eyes provide a wide field of view — an anti-predator adaptation for open environments where cover is sparse. Long-distance threat detection is a primary visual function.
2. What does it sense? — Large ears provide acute directional hearing. Scent glands used for mate recognition and territorial communication. In open habitat with limited canopy, hearing and smell compensate for reduced visual cover.
3. How does it move? — Quadrupedal with an unusual locomotor repertoire: walks slowly, trots at moderate speed, or switches to a galloping gait capable of reaching speeds near 45 km/h. Also performs a stiff-legged pronking (bounding) display, similar to African ungulates. Despite being a rodent, its body plan and gait resemble a small deer or antelope — an example of convergent functional evolution.
4. How does it breathe? — Continuous pulmonary respiration. No specialized respiratory adaptations. Cardiovascular capacity supports burst-speed escape locomotion.
5. What fuels it? — Grasses and shrubs dominate the diet. The Patagonian steppe has limited plant diversity — the Cavy is adapted to extract maximum nutrition from coarse, low-nutrient vegetation. Water is obtained both through plant matter and from direct water sources.
6. What limits it? — Predation by pumas, foxes, and birds of prey. Habitat conversion for sheep farming has reduced available range. Breeding pairs maintain long-term monogamous bonds, and the loss of a mate affects reproductive success significantly.

Listed as Near Threatened by the IUCN.

7. What decision is constant? — Stay in the open or move to cover. Patagonian Cavies navigate a constant trade-off between exposed foraging zones (where food is) and shelter areas (where predators are slower to reach them). Every foraging session involves vigilance.
8. What role does it play? — Grazer and prey species. As a medium-sized herbivore, the Patagonian Cavy is a key link in the Patagonian food web — converting plant material into protein that supports carnivores. Burrows excavated by the species also provide habitat structure for other animals.

Reflection:: The Cavy looks like a small deer but is a rodent. What does this tell you about the relationship between form and family?

Sources:: IUCN Red List — Dolichotis patagonum; Patagonian wildlife field research; primary zoological reference materials; ungulate convergence literature.

TOTEM 6 — THREE-BANDED ARMADILLO

THREE-BANDED ARMADILLO

Tolypeutes matacus

Order: Cingulata

Native to: Central South America — Bolivia, Brazil, Paraguay, Argentina; savanna and scrubland specialist

Diet: Primarily insectivorous — ants, termites, larvae; some plant material

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs, delivered through closed circulatory system

1. What does it see? — Small eyes with limited visual acuity. Vision is not the primary sense for hunting or predator detection in this species. Low-light function is adequate but not exceptional.
2. What does it sense? — Highly developed olfaction. The primary hunting tool — the Three-Banded Armadillo locates subsurface insect colonies by smell and sound vibration through the substrate. Long claws break open termite mounds and ant colonies with precision.
3. How does it move? — Quadrupedal, walking on the tips of its claws. Slow and deliberate in foraging mode. When threatened, it does not run — it rolls into a near-perfect sphere using

its armored plates, head, and tail as an interlocking defensive shell. The ball is nearly impenetrable to most predators.

4. How does it breathe? — Continuous pulmonary respiration. When rolled into a defensive ball, the breathing aperture is minimal but functional. Ventilation continues through the small gap maintained by the interlocking plates.
5. What fuels it? — Insects, primarily ants and termites. A single Three-Banded Armadillo may consume thousands of insects per foraging session. The diet is extremely protein-dense and requires constant foraging to sustain metabolic needs.
6. What limits it? — Habitat loss and hunting. The species is listed as Vulnerable by the IUCN. Its rolling defense is effective against natural predators but does not protect it from human harvest. It cannot outrun threats and relies entirely on passive defense.
7. What decision is constant? — Roll or run. The Three-Banded Armadillo is one of only two armadillo species capable of complete self-enclosure. Every threat assessment ends in one of two responses: continue and tolerate, or roll and wait. There is no middle option.
8. What role does it play? — Insect population regulation. By consuming large volumes of ants and termites, the armadillo helps regulate colony sizes that, left unchecked, could destabilize local soil and plant communities. Its excavation behavior also aerates soil and disturbs substrate in ways that benefit other species.

Reflection:: A defense that works against every predator except humans — what does that tell you about the limits of biological adaptation?

Sources:: IUCN Red List — Tolypeutes matacus; South American wildlife field research; primary zoological reference materials; cingulate biology literature.

TOTEM 7 — TWO-TOED SLOTH

TWO-TOED SLOTH

Choloepus didactylus

Order: Pilosa

Native to: Northern South America — Colombia, Venezuela, Guyana, Ecuador, Peru, Brazil; rainforest canopy specialist

Diet: Herbivorous — leaves dominant; fruit, flowers, and occasionally small vertebrates as secondary sources

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs; unique muscular adaptations for inverted respiration

1. What does it see? — Low-acuity vision. Poor color discrimination. Functional in low light. The Two-Toed Sloth does not rely on vision as its primary interface with the environment.
2. What does it sense? — Touch and proprioception are primary. Long curved claws maintain grip on branches without conscious muscular effort. Olfaction guides food selection. The sloth's relationship with its environment is tactile and chemical more than visual.
3. How does it move? — Inverted suspension locomotion — the sloth hangs upside down and moves through the canopy hand-over-hand. Movement is deliberate and slow. The metabolic strategy of the Two-Toed Sloth is built around minimizing energy expenditure — slow movement is not a failure of speed, it is the entire survival strategy.
4. How does it breathe? — Pulmonary respiration with a critical anatomical adaptation: the internal organs are attached to the ribcage rather than suspended freely, which prevents the weight of the organs from compressing the lungs during inverted hanging. Without this adaptation, the sloth would not be able to breathe while in its natural resting and moving posture.
5. What fuels it? — Leaves are the primary fuel. Leaves are low in calories and difficult to digest. The sloth's digestive system is extremely slow — the gut can take up to a month to process a full load of leaves. The metabolic rate is among the lowest of any mammal. Everything about the Two-Toed Sloth's biology is calibrated to survive on minimal caloric input.
6. What limits it? — Habitat loss and fragmentation are the primary threats. The sloth's dependence on canopy connectivity means that deforestation creates uncrossable gaps in the landscape — the sloth cannot travel through open ground safely or efficiently. Body temperature regulation is also a limit — sloths depend on ambient temperature and behavioral thermoregulation (moving between sun and shade in the canopy).
7. What decision is constant? — Move or stay still. The Two-Toed Sloth spends up to 20 hours per day in stillness. Movement is reserved for feeding and repositioning. Every movement is metabolically costly and must be justified by nutritional gain.
8. What role does it play? — Host ecosystem. The sloth's fur is a unique microhabitat — algae grows in specialized grooves in the fur, providing camouflage (green tint in the canopy) and supporting a community of moths, beetles, and other invertebrates that live exclusively on sloths. The sloth is not just an animal — it is a moving ecosystem.

Reflection:: An animal that barely moves, barely eats, and barely generates heat — yet survives. What does this force you to reconsider about what "successful" looks like in biology?

Sources:: IUCN Red List — Choloepus didactylus; sloth biology research (Jonathan Pauli, Becky Cliffe); primary zoological reference materials; pilosa anatomy literature.

TOTEM 8 — AFRICAN CRESTED PORCUPINE

AFRICAN CRESTED PORCUPINE

Hystrix cristata

Order: Rodentia

Native to: Sub-Saharan Africa, North Africa, and Italy (introduced); wide habitat tolerance — forests, rocky hills, agricultural areas

Diet: Primarily herbivorous — roots, bulbs, bark, fallen fruit, and cultivated crops; occasionally scavenges bone for calcium

Respiration: Continuous pulmonary breathing — oxygen extracted from air via lungs, delivered through closed circulatory system

1. What does it see? — Limited visual acuity. The African Crested Porcupine is primarily nocturnal and does not rely on vision for most behaviors. Eyes function adequately in low light but are not the primary sensory interface.
2. What does it sense? — Acute olfaction — primary tool for food location and predator detection. Hearing is well-developed. Tactile whiskers assist navigation in darkness and in underground environments.
3. How does it move? — Quadrupedal and terrestrial. Walks and runs on all four limbs. When threatened, does not flee immediately — instead, it erects its quills, rattles them (producing a distinctive warning sound by shaking hollow tail quills), and may charge backward toward the threat. Quills are released on contact but are not actively thrown — contact with the predator causes them to embed.
4. How does it breathe? — Continuous pulmonary respiration. No specialized respiratory adaptations. Standard mammalian mechanics.
5. What fuels it? — Roots, bulbs, and bark are primary food sources — underground and ground-level foraging. Dietary calcium is a specific need — porcupines have been documented gnawing bones and visiting mineral licks to supplement calcium for quill production and skeletal maintenance.
6. What limits it? — Predation by large carnivores (lions, leopards, hyenas) — though quills make predation difficult and costly. Persecution by humans due to crop raiding limits

populations in agricultural zones. Quill density is finite — the animal must balance quill investment against nutritional resources.

7. What decision is constant? — Defend or move. The African Crested Porcupine does not run as its first response to threat. Its defense system requires the animal to stand its ground. This is a commitment — the defense only works if the animal holds position. Movement is calculated.
8. What role does it play? — Soil aerator and seed disperser. Extensive burrowing and root-digging disrupts and aerates soil. Fruit and seed consumption, followed by movement across home ranges, disperses seeds at distance from parent plants.

Reflection:: The quills cannot be thrown. The defense only works if the predator makes contact. What does it mean to have a defense that requires the threat to arrive?

Sources:: IUCN Red List — Hystrix cristata; African mammal field guides; primary zoological reference materials; hystricid biology literature.

OUTSIDE ANIMAL TOTEMS — BATCH 1

WARTHOG

Phacochoerus africanus

Order: Artiodactyla

Native to: Sub-Saharan Africa

Diet: Omnivore (grasses, roots, bark, insects)

Respiration: Lungs

1. What does it see?

Warthogs have eyes positioned high and laterally on the skull, providing a wide field of view that allows detection of predators from nearly all directions while grazing with the head down. Their color vision is limited compared to primates, but motion detection in peripheral vision is acute.

2. What does it sense?

The warthog's large, mobile ears rotate independently to localize sounds, detecting approach from predators even in tall grass. A well-developed sense of smell is used to locate underground food sources and to read chemical signals left by other warthogs.

3. How does it move?

Warthogs move on their wrist pads when rooting and foraging, bearing weight on the carpal joints rather than the hooves. When fleeing predators, they can sustain speeds up to 55 km/h, holding the tail erect as a visible signal to other warthogs in the group.

4. How does it breathe?

Lungs draw air through the nostrils, which are reinforced by the cartilaginous warthog snout built for rooting in compacted soil. At high locomotor speeds, respiratory rate increases sharply to meet the elevated oxygen demand of the running muscles.

5. What fuels it?

Grasses and grass rhizomes form the bulk of the diet in the wet season, while roots, bulbs, and bark become primary energy sources during dry months when grasses die back. Insects and carrion are consumed opportunistically, contributing protein when plant material is nutritionally lean.

6. What limits it?

Body size and limited insulating fat make warthogs vulnerable to cold; they shelter in burrows (often those excavated by aardvarks) to retain heat overnight. Dental wear sets a functional limit on lifespan, as the large incisors and molars used for root-digging erode over years of abrasive soil contact.

7. What decision is constant?

When does remaining near the foraging patch outweigh the growing risk of predator detection in open ground? When is the burrow entrance far enough away that retreating to it costs more energy than it protects?

8. What role does it play?

Warthogs disturb and aerate soil through continuous rooting, exposing seeds and invertebrates that become available to other species. Their carcasses and dung supply nutrients that cycle through the savanna system, and they serve as prey that sustains large African carnivores including lions and leopards.

Reflection

What does the warthog's body tell you about the tradeoffs between finding food underground and staying alert above it?

Sources (Internal Only)

— Park observation records

— Species care protocols

DOMESTIC PIG

Sus scrofa domesticus

Order: Artiodactyla

Native to: Eurasia (domesticated worldwide)

Diet: Omnivore (grains, vegetation, roots)

Respiration: Lungs

1. What does it see?

Pigs have a visual field of approximately 310 degrees due to laterally placed eyes, providing broad detection of movement around them while limiting binocular depth perception to a narrow frontal zone. They possess dichromatic color vision and rely heavily on motion cues rather than fine detail.

2. What does it sense?

The pig's disc-shaped snout contains a dense concentration of tactile mechanoreceptors (Eimer's organs) that detect texture, temperature, and soil composition during rooting. Olfactory acuity is exceptional; pigs can detect odors several meters underground and use scent to identify individuals, reproductive state, and feed quality.

3. How does it move?

Pigs walk on two of their four toes (digits three and four), with the lateral digits making only light contact with the ground on soft surfaces. Despite a heavy-bodied appearance, they can run at speeds exceeding 18 km/h and are capable swimmers when motivated.

4. How does it breathe?

Lungs expand and contract via a muscular diaphragm; the snout acts as the primary air-intake structure, filtering particles during rooting. Thermoregulation is constrained because pigs lack functional sweat glands; panting and wallowing in mud or water serve as the primary mechanisms for heat dissipation.

5. What fuels it?

A digestive system similar to the human monogastric gut allows pigs to extract energy efficiently from starchy grains and diverse plant material. A handful of pellets and root vegetables supply the carbohydrate base, while protein requirements are met through legume meal, insects, or animal byproducts in managed diets.

6. What limits it?

The absence of sweat glands creates a hard thermal ceiling; body temperature rises rapidly in heat without access to mud, shade, or water for wallowing. Rooting behavior, while

metabolically productive, is energetically expensive and becomes restricted when ground freezes or hardens.

7. What decision is constant?

When does the energy cost of rooting deeper into harder substrate exceed the caloric value of what is likely to be found? When does proximity to other pigs represent competition for a resource versus shared vigilance against disturbance?

8. What role does it play?

In semi-feral and feral contexts, pigs act as ecosystem engineers by turning soil through rooting, which alters plant community composition and creates microhabitats used by invertebrates and small vertebrates. Fecal deposition distributes seeds and nutrients across the landscape, cycling organic matter through the soil system.

Reflection

What does the pig's snout reveal about the relationship between sensory investment and how an animal earns its living?

Sources (Internal Only)

— Park observation records

— Species care protocols

DOMESTIC GOAT

Capra aegagrus hircus

Order: Artiodactyla

Native to: Western Asia (domesticated worldwide)

Diet: Herbivore (browse, grasses, bark)

Respiration: Lungs

1. What does it see?

Goats have rectangular, horizontal pupils that expand their visual field to nearly 320 degrees, enabling detection of predator movement across open terrain while the head is lowered to feed. This pupil shape provides a wide depth of field across the horizontal plane, keeping more of the visual environment in focus simultaneously.

2. What does it sense?

The goat's upper lip is split and highly mobile, functioning as a tactile organ that sorts plant material by texture and moisture before it reaches the molars. Scent glands located below the

eyes (preorbital), on the feet (interdigital), and near the tail produce chemical signals used in individual recognition and reproductive communication.

3. How does it move?

Hooves are split into two toes with hard outer walls and softer inner pads that grip irregular surfaces, allowing goats to stand on rocky slopes and narrow ledges. The skeletal structure of the carpals and tarsals allows independent articulation of each digit, enabling precise foot placement on uneven terrain.

4. How does it breathe?

Air moves through the lungs via a diaphragm-driven bellows system; the respiratory rate increases during exertion on steep terrain. Goats do not pant as a primary cooling strategy; they use shade-seeking and reduced activity during peak heat, supplemented by evaporative cooling through skin and respiratory surfaces.

5. What fuels it?

As ruminants, goats ferment plant material in a four-chambered stomach, extracting nutrients from fibrous browse, bark, and coarse grasses that monogastric animals cannot efficiently process. Mulberry leaves, local browse, and woody stems are preferred over uniform pasture grasses, reflecting an evolutionary history in scrubby, varied terrain.

6. What limits it?

Copper sensitivity creates a narrow dietary tolerance; excess copper in feed quickly accumulates to toxic liver levels, making mineral balance a continuous constraint in managed care. Wet conditions sustained over time predispose hooves to rot and foot infections, limiting activity and food intake on waterlogged ground.

7. What decision is constant?

When does the energy available from the current browse patch justify continued stripping of bark, and when does moving to a new stem become more efficient? When does the height advantage of climbing a structure produce enough new forage to offset the energy of the climb?

8. What role does it play?

In natural and feral settings, goats function as browsers that suppress woody shrub encroachment, opening habitat structure for grasses and low-growing plants used by other species. Selective removal of plant tissue influences canopy composition and light penetration at the ground layer.

Reflection

What does a goat's pupil shape tell you about the environment it evolved to survive in?

Sources (Internal Only)

— Park observation records

— Species care protocols

DOMESTIC SHEEP

Ovis aries

Order: Artiodactyla

Native to: Western Asia (domesticated worldwide)

Diet: Herbivore (grasses, forbs)

Respiration: Lungs

1. What does it see?

Sheep possess wide-set eyes with rectangular horizontal pupils, producing a visual field of approximately 270–320 degrees that enables near-panoramic surveillance for predators without head movement. Their binocular zone is narrow but sufficient for judging distances to footing obstacles directly in front of the muzzle.

2. What does it sense?

The sheep's vomeronasal organ (Jacobson's organ) in the roof of the mouth detects non-volatile chemical compounds, particularly those in urine and glandular secretions, and plays a central role in reproductive signaling between rams and ewes. Hearing is acute; the mobile ear pinnae rotate toward sound sources to pinpoint direction and distance of disturbances.

3. How does it move?

Sheep move on cloven hooves with hard keratin walls and softer digital pads, providing grip on uneven pasture ground. When alarmed, the flock synchronizes direction and speed simultaneously — a behavior driven by visual contagion where one individual's flight posture triggers immediate matching movement in neighbors.

4. How does it breathe?

Resting respiration in sheep averages 12–20 breaths per minute, rising sharply during flight and thermal stress. Evaporative water loss through the respiratory tract is the primary cooling mechanism during heat; wool insulation, while thermally protective in cold, creates heat retention challenges in warm climates without shearing.

5. What fuels it?

Sheep are ruminants with a four-chambered stomach; ingested grasses and forbs are fermented by microbial populations in the rumen before the resulting volatile fatty acids are absorbed as the primary energy source. A handful of hay or pasture grass per feeding session supplies the fibrous substrate that keeps rumen fermentation active and gut motility regular.

6. What limits it?

Internal parasites — particularly gastrointestinal nematodes — are a consistent physiological constraint, as sheep have limited behavioral avoidance of contaminated pasture and show lower natural resistance than goats. Wool growth continues independently of season and will mat, accumulate fecal debris, and create heat and parasite risk if not periodically removed.

7. What decision is constant?

When does the safety of remaining with the flock outweigh moving toward a more productive patch of forage separated from the group? When does the time spent at the current grazing site produce diminishing returns relative to the risk of moving to unfamiliar ground?

8. What role does it play?

Grazing sheep selectively crop grasses and forbs to ground level, promoting growth of low-growing plants and preventing tall grass species from outcompeting them. Hoof traffic and dung deposition together alter soil structure and nutrient cycling across grazed landscapes.

Reflection

What advantage does a prey animal gain from having its eyes on the sides of its head instead of the front?

Sources (Internal Only)

— Park observation records

— Species care protocols

ALPACA

Vicugna pacos

Order: Artiodactyla

Native to: South America (Andes)

Diet: Herbivore (grasses, hay)

Respiration: Lungs

1. What does it see?

Alpacas have laterally placed eyes with a wide visual field exceeding 300 degrees, adapted to detecting movement across open Andean plateaus where visual horizons are broad and predators approach from distance. The pupils are elongated horizontally, expanding peripheral depth of field while the animal grazes with its head lowered.

2. What does it sense?

Alpacas use a highly mobile upper lip to sort forage by texture before ingestion, and they investigate unfamiliar objects primarily through close olfactory inspection. Ear movement is pronounced and independent; rotating pinnae allow localization of sound sources, and ear position communicates social status and alertness state within the herd.

3. How does it move?

Alpacas move on padded two-toed feet; unlike sheep and goats, their feet have a leathery sole pad rather than hard hooves, which causes less soil disturbance but limits traction on steep wet rock. They walk and trot using a lateral gait (moving both limbs on the same side together), which is efficient for sustained travel over high-altitude uneven terrain.

4. How does it breathe?

Alpacas evolved at altitudes of 3,500–5,000 meters and have red blood cells with an unusually high hemoglobin oxygen affinity, allowing efficient oxygen extraction from thin, high-altitude air. At lower elevations, this same physiology results in exceptional respiratory efficiency during exertion, with respiratory rates recovering quickly after movement.

5. What fuels it?

As modified ruminants (camelids have a three-compartment stomach rather than four), alpacas ferment plant material highly efficiently and extract more nutrition per unit of forage than sheep or goats, making them well-suited to sparse, low-quality Andean grasses and hay. A handful of grass hay twice daily provides the fiber base needed for rumen fermentation, supplemented by mineral access.

6. What limits it?

Dental malocclusion is a recurring constraint; alpaca incisors grow continuously and misalignment reduces foraging efficiency over time, requiring managed trimming in captive settings. Heat stress is a significant risk at lower altitudes, as the dense fiber coat adapted for Andean cold retains heat in warm humid environments without shade and water access.

7. What decision is constant?

When does the proximity of an unfamiliar animal or object cross a threshold that triggers alarm calling and group movement away? When does returning to a depleted patch of hay represent less energy cost than competing for a fuller one occupied by a dominant herd member?

8. What role does it play?

In Andean grassland ecosystems, alpacas and their wild ancestor the vicuña graze selectively in ways that can maintain grass diversity if stocking density stays within carrying capacity. Dung is concentrated in communal latrine areas, creating localized nutrient hotspots that differ chemically from surrounding grazed soil.

Reflection

What does the alpaca's blood chemistry tell you about how the body adapts to extreme environments over generations?

Sources (Internal Only)

— Park observation records

— Species care protocols

EMU

Dromaius novaehollandiae

Order: Casuariiformes

Native to: Australia

Diet: Omnivore (seeds, fruit, insects, plants)

Respiration: Lungs

1. What does it see?

Emus have large eyes positioned on the sides of the head, providing a wide monocular field on each side and a functional forward binocular zone used when approaching and inspecting food items at close range. Vision is acute in bright daylight conditions, and emus can detect slow movement at long distance across flat open terrain.

2. What does it sense?

Emus possess an inflatable neck sac that produces low-frequency booming calls used in long-distance communication; the vibrational component of these calls may transmit through the ground and be detected by other emus at range. Their sense of smell is functional but not dominant; food detection relies primarily on vision, with close inspection of items before ingestion.

3. How does it move?

Emus are ratites with reduced wings and a three-toed foot bearing a powerful inner claw; they are bipedal runners capable of sustained speeds of 48–50 km/h using a striding gait driven by large hip and thigh muscles. The emu's legs carry no feathers below the thigh, reducing thermal load during the prolonged walking and trotting used to cover large daily distances in

search of food.

4. How does it breathe?

Air passes through the lungs in a flow-through system (as in all birds), with air sacs serving as bellows that push fresh air continuously in one direction over the gas-exchange surface rather than moving in and out of a dead-end lung. During heat, emus use panting — rapid shallow breathing that increases evaporative cooling from the moist surfaces of the respiratory tract.

5. What fuels it?

Emus are opportunistic feeders that shift diet composition seasonally: seeds and fruit dominate when available, while insects (particularly caterpillars and grasshoppers) supply concentrated protein during breeding and chick-rearing periods. One handful of pellets paired with access to seeds, fruit pieces, and live insects covers the energy and protein demands of a captive emu in a day.

6. What limits it?

Emus cannot fly, making escape from enclosed areas impossible once contained; this behavioral inflexibility is managed by the long-distance nomadism that characterizes wild population movement. The proventricular portion of the digestive system is adapted for grit-assisted mechanical breakdown of hard seeds, and lack of access to grit or stones impairs digestion in captive settings.

7. What decision is constant?

When does the energy content of seeds remaining in the current area justify continued searching, and when does the distance to the next known resource patch become worth traveling? When does an unusual sound or shape in the environment require approach and investigation versus immediate withdrawal?

8. What role does it play?

Emus are long-distance seed dispersers; seeds consumed whole pass intact through the gut and are deposited far from the parent plant in dung that also fertilizes the germination site. Their movements across hundreds of kilometers track rainfall and resource pulses, redistributing plant propagules across the Australian landscape.

Reflection

What does the emu's inability to fly, combined with its capacity to run dozens of kilometers in a day, reveal about how energy is allocated differently across bird species?

Sources (Internal Only)

— Park observation records

— Species care protocols

OSTRICH

Struthio camelus

Order: Struthioniformes

Native to: Africa

Diet: Omnivore (plants, seeds, insects, small vertebrates)

Respiration: Lungs

1. What does it see?

The ostrich has the largest eyes of any living terrestrial vertebrate — each eye measures approximately 5 cm in diameter — providing exceptional acuity and the ability to detect moving objects at distances exceeding 3.5 km across open savanna. Eyes are positioned laterally, but the bird can swivel its long neck to bring either eye to bear on a target with precision.

2. What does it sense?

Hearing is acute; the large, mobile ear openings (lacking external pinnae) funnel sound directly to the tympanic membrane, enabling detection of low-frequency calls from other ostriches at long range. Tactile sensitivity in the bill tip allows discrimination of seed size, texture, and surface hardness during foraging on mixed ground substrate.

3. How does it move?

Ostriches are the fastest running birds on Earth, capable of sustained speeds of 45–50 km/h and short bursts exceeding 70 km/h, powered by large leg muscles anchored to a reduced but massive pelvic girdle. They are the only birds with two toes; the large inner toe bears the full weight and carries a blunt nail used as a defensive weapon capable of delivering fatal kicks.

4. How does it breathe?

Ostriches use the avian flow-through respiratory system with air sacs acting as bellows; at running speed, each stride compresses and expands the air sac system, creating mechanical coupling between locomotion and ventilation that reduces the energy cost of breathing during sustained running. During heat, they raise their feathers to expose the less-insulated skin of the inner thigh, increasing radiative and convective heat loss.

5. What fuels it?

Ostriches consume a wide variety of plant parts — seeds, succulents, grasses — along with insects, lizards, and small vertebrates when encountered; no food item is actively stalked but opportunistically taken during movement. A handful of pellets plus access to grasses, a few seeds, and occasional animal protein pieces supports the daily energy budget of a captive ostrich.

6. What limits it?

Ostriches swallow grit and stones deliberately to assist the gizzard in mechanically grinding hard seeds; without access to grit, digestive efficiency drops substantially. Their reproductive cycle is tightly coupled to thermoregulation: nest incubation on open ground exposes both eggs and incubating adults to sustained heat loading, which limits nesting success in extreme temperatures.

7. What decision is constant?

When does the combined vigilance of multiple ostriches in a group reduce individual predator monitoring enough to allow longer uninterrupted feeding bouts? When does the approach of an unfamiliar large animal require a kick, a display, or flight as the lower-cost response?

8. What role does it play?

Ostriches are bulk seed dispersers and disturb soil through scratching and incidental digging, creating depressions that collect water and support localized plant germination after rains. As large-bodied prey, they sustain populations of lions, cheetahs, and spotted hyenas that regulate the broader herbivore community.

Reflection

What does having the largest eye of any land animal — paired with a brain smaller than either eye — tell you about where the ostrich's survival investment is concentrated?

Sources (Internal Only)

— Park observation records

— Species care protocols

DROMEDARY CAMEL

Camelus dromedarius

Order: Artiodactyla

Native to: Arabian Peninsula / North Africa

Diet: Herbivore (thorny plants, dry grasses, browse)

Respiration: Lungs

1. What does it see?

Camels have a wide lateral visual field due to eye placement on the sides of the head, supplemented by a third eyelid (nictitating membrane) that is transparent enough to allow vision during sandstorms while protecting the corneal surface. Long double-row eyelashes

reduce glare from bright reflective desert surfaces and physically block windblown sand from reaching the eye.

2. What does it sense?

The camel's nostrils can seal completely — closing to a narrow slit — during sandstorms to exclude particulates while still allowing passive gas exchange through the nasal mucosa. Olfaction is the primary tool for evaluating forage quality and reproductive state; camels perform flehmen (curling the upper lip and drawing air across the vomeronasal organ) to read chemical signals from urine, dung, and secretions.

3. How does it move?

Camels are pacers: both legs on the same side move forward together, producing a rolling gait that is energetically efficient for sustained travel over loose sand and rocky desert substrate. The two broad, leathery toes spread under body weight to distribute load across a wide surface area, preventing the foot from sinking into soft sand.

4. How does it breathe?

Camel nasal passages are structurally convoluted (turbinate-rich), recapturing up to 66% of the moisture from exhaled air before it exits the nostril — a critical mechanism for reducing respiratory water loss in arid environments. Resting metabolic rate is slightly lower than expected for body size, reducing both water loss and heat production during periods of inactivity in heat.

5. What fuels it?

The hump stores fat — not water — as a concentrated energy reserve that can be metabolized during food scarcity; fat catabolism also produces metabolic water as a byproduct of oxidation. Thorny plants, dry browse, and salt-tolerant succulents that other ungulates avoid form the dietary base; camels can consume halophytic plants with salt concentrations toxic to most mammals and excrete excess salt through highly concentrated urine.

6. What limits it?

Prolonged water deprivation beyond body condition loss of ~25% of body mass begins to impair renal function; camels can tolerate this level of dehydration and rehydrate rapidly (consuming up to 135 liters in minutes) but cannot survive indefinitely without water access. Hump depletion from extended food deprivation causes the hump to soften and lean to the side, indicating negative energy balance that impairs thermoregulation and locomotor performance.

7. What decision is constant?

When does the energy cost of walking to the next known water source become less than the cost of remaining in the current location without water? When does consuming a salt-laden

plant represent a net gain or loss given the additional water required to process and excrete the salt load?

8. What role does it play?

In arid ecosystems, camels are one of few large herbivores capable of processing thorny, xeric, and salt-tolerant vegetation that remains unavailable to other grazers, and their selective consumption helps shape the structure of arid plant communities. Dung deposits in desert environments provide organic matter that supports specialized invertebrate and microbial communities in otherwise nutrient-sparse soils.

Reflection

What does the camel's body tell you about how much biological machinery is required to survive in an environment where water is the limiting resource?

Sources (Internal Only)

— Park observation records

— Species care protocols

DONKEY

Equus asinus

Order: Perissodactyla

Native to: Northeast Africa

Diet: Herbivore (grasses, browse, hay)

Respiration: Lungs

1. What does it see?

Donkeys have large eyes set wide on the sides of the head, giving a visual field close to 350 degrees with only a narrow blind spot directly behind and immediately in front of the nose. This near-panoramic field is an adaptation to open desert terrain where predators may approach from any direction; the broad binocular overlap is narrower than in forward-facing predators, trading depth perception for coverage.

2. What does it sense?

The donkey's large, long ears can rotate independently through a wide arc, pinpointing the direction of sounds with high accuracy and detecting sounds at greater distances than horses. Tactile sensitivity in the muzzle and lips allows precise selection of individual plant stems, enabling donkeys to forage on sparse, patchy vegetation by picking through substrate at fine resolution.

3. How does it move?

Donkeys are single-toed ungulates moving on a single enlarged digit (digit III) encased in a hard hoof; the hoof wall and frog (the V-shaped pad on the underside) together absorb impact and provide grip. Donkeys are more sure-footed on rocky terrain than horses due to a narrower, more upright hoof shape and a tendency to place each foot with deliberate evaluation before committing full weight — a trait tied to cautious behavioral responses rather than impulsivity.

4. How does it breathe?

Lungs are large relative to body mass; donkeys originated in arid desert ecosystems where thermoregulation through evaporative sweating is critical, and they sweat more efficiently than camels but less copiously than horses. Respiratory rate increases during exertion and during heat loading; the large nasal passages allow high-volume airflow during sustained work.

5. What fuels it?

The donkey's hindgut fermentation system (cecum and large colon) is highly efficient at extracting nutrients from coarse, low-quality forage that horses and ruminants underutilize; this allows donkeys to maintain body condition on sparse desert browse and dry grasses. A handful of low-sugar grass hay provides the fibrous substrate needed to keep hindgut microbial fermentation active; access to mulberry leaves and local browse adds variety consistent with the broad diet of wild ancestors.

6. What limits it?

Donkeys are prone to hyperlipaemia — a dangerous mobilization of fat into the bloodstream — when feed is suddenly restricted or withheld; this makes fasting or abrupt diet changes a metabolic risk rather than a neutral management strategy. Hoof overgrowth in captive settings without abrasive terrain contact requires regular trimming to prevent structural deformity and lameness.

7. What decision is constant?

When does an unfamiliar object or sound warrant standing still and assessing (a common donkey response) rather than fleeing, and what sensory threshold determines which response is triggered? When does the distance to the next forage patch on sparse ground justify the energy expenditure of travel versus continued extraction from the current site?

8. What role does it play?

In their native northeast African range, wild asses browse on drought-resistant shrubs and grasses in rocky desert terrain, preventing any single plant species from monopolizing patchy water-adjacent areas. As prey for lions and leopards in historical ranges, they contributed to the energy transfer between primary producers and apex carnivores in arid-land food webs.

Reflection

What does the donkey's tendency to stop and assess rather than flee immediately tell you about how different prey animals have evolved different risk-management strategies?

Sources (Internal Only)

— Park observation records

— Species care protocols

HORSE

Equus caballus

Order: Perissodactyla

Native to: Central Asia (domesticated worldwide)

Diet: Herbivore (grasses, hay)

Respiration: Lungs

1. What does it see?

Horses have the largest eyes of any terrestrial mammal relative to skull size, positioned laterally to produce a near-360-degree visual field with binocular vision in a frontal zone of approximately 65 degrees used for judging distances during locomotion. They can see in dim light effectively due to a reflective tapetum lucidum behind the retina but have two blind spots: directly behind the hindquarters and within approximately 1 meter directly in front of the muzzle.

2. What does it sense?

Hearing is highly acute; the large mobile ear pinnae can rotate up to 180 degrees independently, with each ear tracking separate sound sources simultaneously. Vibrissae (whiskers) around the muzzle and eyes are mechanosensory organs that detect air movement and provide tactile information in the visual blind spot close to the face — removal of vibrissae in groomed horses eliminates this sensory channel entirely.

3. How does it move?

Horses move on a single digit — the third toe — encased in a hoof made of highly keratinized tissue; the internal digital cushion and the frog absorb concussive forces that would otherwise damage bone and tendon during sustained galloping. At the gallop, horses use a "locomotion-ventilation coupling" mechanism where each stride compresses and expands the thorax, mechanically driving a breath cycle that reduces the additional muscular cost of breathing at speed.

4. How does it breathe?

Horses are obligate nasal breathers; the soft palate forms a near-permanent seal with the epiglottis that routes air exclusively through the nasal passages and prevents mouth breathing at rest and during most exertion. Thermoregulation depends heavily on sweating; horse sweat is rich in latherin, a surfactant protein that reduces sweat surface tension and allows efficient spreading across the dense coat for evaporative cooling.

5. What fuels it?

The horse's hindgut — particularly the cecum and ventral colon — houses microbial populations that ferment fibrous plant cell walls into volatile fatty acids, the primary energy currency for the animal. A handful of grass hay at multiple feeding intervals throughout the day supports continuous hindgut fermentation; the system is designed for near-continuous low-volume intake rather than periodic large meals.

6. What limits it?

The anatomy that enables near-continuous grazing (large, hypsodont molars that grind laterally) also creates a constraint: dental surfaces wear at a predictable rate over a lifetime, and horses with severely worn molars in old age lose the ability to process fibrous forage efficiently, leading to body condition decline. Colic — disruption of gut motility — is a primary cause of morbidity and mortality in horses, triggered by diet changes, dehydration, sand ingestion, or prolonged lack of movement.

7. What decision is constant?

When does the alarm signal from one herd member carry enough information to trigger full flight, versus an ambiguous signal that results in group orientation and attention without movement? When does continued grazing at a productive grass patch become less favorable than moving with the group to maintain herd cohesion?

8. What role does it play?

In grassland ecosystems, horses are bulk grazers that crop vegetation to a consistent height, promoting grass tillering and preventing the accumulation of dead material that suppresses new growth. Their hindgut fermentation produces dung with intact, viable seeds, making them long-distance seed dispersers; dung piles also serve as nutrient-deposition points that support distinct plant and invertebrate assemblages.

Reflection

What does the horse's dependence on continuous forage intake — rather than periodic large meals — reveal about the design constraints of a fermentation-based digestive system?

Sources (Internal Only)

— Park observation records

— Species care protocols

NORTH AMERICAN RIVER OTTER

Lontra canadensis

Order: Carnivora

Native to: North America

Diet: Carnivore (fish, crayfish, amphibians)

Respiration: Lungs

1. What does it see?

River otters have eyes adapted for both aerial and underwater use; a flat cornea reduces the optical distortion that occurs when light transitions from water to the lens, providing functional acuity in both media. Underwater vision is sufficient to track actively moving fish at close range, and the otter's pupils constrict to a point in bright surface light to reduce glare during surface scanning.

2. What does it sense?

Whiskers (vibrissae) are long, thick, and highly innervated; they detect water pressure waves generated by the movement of fish or crayfish in murky or dark water, functioning as a hydrodynamic detection system that operates independently of vision. Scent marking with anal gland secretions deposited at latrines on the water's edge communicates individual identity, territory use, and reproductive condition to other otters moving through the area.

3. How does it move?

In water, the otter propels itself primarily by dorsoventral undulation of the spine and tail, with the webbed hind feet functioning as rudders during turns; forelimbs are held against the body during fast swimming. On land, otters move with a bounding gait using all four limbs, often tobogganing on snow or mud banks by stretching the body flat and sliding — a behavior that also reduces the energy cost of downslope travel.

4. How does it breathe?

As a fully air-breathing mammal, the otter surfaces regularly during foraging dives; typical dives last 30–60 seconds, with a physiological capacity for dives extending to several minutes. During submersion, heart rate drops through a diving reflex that reduces peripheral circulation and conserves oxygen for the brain and heart — a mechanism shared across diving mammals and birds.

5. What fuels it?

Fish are the primary prey, seized with the teeth after an underwater pursuit and consumed at the surface or on a bank; crayfish and amphibians supplement the diet seasonally, particularly when fish are less available in winter-cold water or during breeding aggregations. One to several whole fish or an equivalent portion of fish pieces alongside crayfish constitutes a daily ration; prey size is calibrated to what can be captured and processed efficiently without tools.

6. What limits it?

Water quality is a direct constraint: otter population density tracks fish abundance and water clarity, and chemical contamination that reduces prey populations removes the energy base that supports otter territories. Fur must be kept clean and groomed; the insulating capacity of otter fur depends on a layer of trapped air within the dense underfur, which is lost if the fur becomes contaminated with oil or sediment, rapidly leading to hypothermia in cold water.

7. What decision is constant?

When does the energetic return from pursuing a fleeing fish through open water exceed the cost of the chase, versus abandoning and searching for a less mobile prey item in substrate cover? When does the current latrine site carry enough chemical information about recent otter activity to make extended investigation worthwhile before moving on?

8. What role does it play?

River otters are apex predators within stream and lake food webs; their selective predation on abundant, easily caught fish species (often slower or injured individuals) creates pressure that shapes fish population age structure and behavior. As consumers of crayfish, otters regulate crayfish population density, indirectly influencing the aquatic vegetation and invertebrate communities that crayfish overgraze when uncontrolled.

Reflection

What does the otter's use of pressure waves through its whiskers tell you about how an animal can gather information in an environment where vision is not the most reliable sense?

Sources (Internal Only)

— Park observation records

— Species care protocols

PART 20

NARRATION SUITES: COMPLETE SYSTEM

The LIFE education system includes 20 narration suites deployed across the facility. Each suite is a themed room designed to move visitors through a progression of concepts — from perception and signal to systems, energy, adaptation, and synthesis.

Narration suites are not decorative. They are structural. Each suite anchors a single principle from the LIFE framework and delivers it through symbols, patterns, and observation. There are no character mascots. There are no text-heavy walls. The room itself is the lesson.

Staff role:: Do not modify narration suite content. Do not add or remove items from narration suites. Do not rearrange furniture, props, or displays. Report any damage immediately.

Authority:: All narration suite content is designed and approved by the owner. No staff member creates, modifies, or themes a narration suite without written owner authorization.

Suite Architecture

Each suite has:

- A title — the suite name
- A governing statement — one line that captures the principle
- A visitor prompt — a short description that frames the exploration
- A QR entry point — SCAN TO ENTER link for the digital layer
- LIFE branding — Language, Intelligence, Form, Ecology

The 20 Narration Suites

Suite 1 — Reflection Suite

Governing Statement:: Perception does not equal reality.

Visitor Prompt:: Explore how our minds construct reality — and where they go wrong.

QR Entry:: https://life.aquarium.com/suite/1

LIFE Principle:: The gap between what you perceive and what exists is where error lives. This suite forces visitors to confront the difference between seeing and understanding.

Suite 2 — Perception Responsibility Suite

Governing Statement:: Ignoring signals causes failure.

Visitor Prompt:: Learn why tuning out warning signs leads systems — and people — to collapse.

QR Entry:: https://life.aquarium.com/suite/2

LIFE Principle:: Every system depends on signal detection. When signals are ignored — in ecosystems, in organizations, in individual behavior — failure follows. Responsibility begins with paying attention.

Suite 3 — Illusion Suite

Governing Statement:: Signal does not equal substance.

Visitor Prompt:: Discover how appearances deceive and how organisms exploit that gap.

QR Entry:: https://life.aquarium.com/suite/3

LIFE Principle:: Mimicry, camouflage, and deception are survival strategies found across every kingdom of life. What looks real is not always real. What looks safe is not always safe.

Suite 4 — Signal Suite

Governing Statement:: Survival depends on detection.

Visitor Prompt:: See how every living system depends on sensing the right signals.

QR Entry:: https://life.aquarium.com/suite/4

LIFE Principle:: From a cell detecting a chemical gradient to a predator reading the movement of prey, signal detection is the foundation of all biological decision-making.

Suite 5 — Signal Reliability Suite

Governing Statement:: False signals break systems.

Visitor Prompt:: Understand why misinformation, mimicry, and noise are existential threats.

QR Entry:: https://life.aquarium.com/suite/5

LIFE Principle:: A system that cannot distinguish true signals from false ones is a system that will eventually fail. This applies to ecosystems, to organizations, and to human judgment.

Suite 6 — Naming Suite

Governing Statement:: Knowledge begins with naming.

Visitor Prompt:: Experience how classification and language unlock understanding of life.

QR Entry:: https://life.aquarium.com/suite/6

LIFE Principle:: You cannot study what you cannot name. Taxonomy, language precision, and naming conventions are the entry point to all structured knowledge.

Suite 7 — Systems Suite

Governing Statement:: Roles create stability.

Visitor Prompt:: Witness how defined roles within ecosystems generate lasting order.

QR Entry:: https://life.aquarium.com/suite/7

LIFE Principle:: Every stable system — biological, organizational, mechanical — depends on defined roles. When roles are clear, the system holds. When roles blur, the system drifts.

Suite 8 — Cooperation Suite

Governing Statement:: Collective strength.

Visitor Prompt:: See how cooperation between organisms creates power no individual has alone.

QR Entry:: https://life.aquarium.com/suite/8

LIFE Principle:: Mutualism, symbiosis, and cooperative behavior are not acts of generosity. They are survival strategies that emerge when the combined output exceeds what any individual could produce.

Suite 9 — Pattern Suite

Governing Statement:: Optimal ranges.

Visitor Prompt:: Explore the narrow bands of conditions where life thrives.

QR Entry:: https://life.aquarium.com/suite/9

LIFE Principle:: Every organism operates within tolerances — temperature, pH, salinity, light. Move outside the band, and the organism fails. Patterns reveal where the band is.

Suite 10 — Constraint Suite

Governing Statement:: Limits shape behavior.

Visitor Prompt:: Discover how boundaries — physical, chemical, biological — define possibilities.

QR Entry:: https://life.aquarium.com/suite/10

LIFE Principle:: Constraints are not obstacles. They are the rules that determine what is possible. Every adaptation is a response to a constraint. Understanding the constraint reveals the design.

Suite 11 — Resource Suite

Governing Statement:: Risk vs. reward.

Visitor Prompt:: Understand the constant calculations every organism makes to survive.

QR Entry:: https://life.aquarium.com/suite/11

LIFE Principle:: Every foraging decision, every territorial defense, every mating display is a cost-benefit analysis. The organism that calculates correctly survives. The one that does not, does not.

Suite 12 — Energy Economy Suite

Governing Statement:: Efficiency vs. speed.

Visitor Prompt:: See the universal trade-off between moving fast and conserving energy.

QR Entry:: https://life.aquarium.com/suite/12

LIFE Principle:: Energy is finite. Every organism must allocate it — to growth, to reproduction, to movement, to defense. The allocation strategy is the organism's identity.

Suite 13 — Adaptation Suite

Governing Statement:: Environment shapes identity.

Visitor Prompt:: Learn how ecosystems sculpt organisms over generations.

QR Entry:: https://life.aquarium.com/suite/13

LIFE Principle:: Form follows function follows environment. Every physical feature of an organism exists because the environment selected for it. Change the environment, and the organism either adapts or disappears.

Suite 14 — Vertical Perspective Suite

Governing Statement:: Height changes awareness.

Visitor Prompt:: Experience how elevation transforms what an organism can perceive and do.

QR Entry:: https://life.aquarium.com/suite/14

LIFE Principle:: A giraffe and a tortoise standing in the same field inhabit different perceptual worlds. Vertical position determines what you see, what you reach, and what threatens you.

Suite 15 — Scale Suite

Governing Statement:: Size changes meaning.

Visitor Prompt:: Discover how the rules of life shift radically across scales of size.

QR Entry:: https://life.aquarium.com/suite/15

LIFE Principle:: Surface tension dominates the world of insects. Gravity dominates the world of elephants. The same physical laws produce completely different realities at different scales.

Suite 16 — Transformation Suite

Governing Statement:: Appearance does not equal function.

Visitor Prompt:: Explore how the same form can serve entirely different purposes.

QR Entry:: https://life.aquarium.com/suite/16

LIFE Principle:: A wing can fly, insulate, display, or balance. A leg can walk, dig, swim, or signal. Form is constant. Function is context-dependent.

Suite 17 — Fragility Suite

Governing Statement:: Environment mismatch.

Visitor Prompt:: Understand why specialized life is both powerful and precarious.

QR Entry:: https://life.aquarium.com/suite/17

LIFE Principle:: Specialization makes an organism powerful within its niche and vulnerable outside it. The more perfectly adapted, the more dependent on the conditions that shaped it.

Suite 18 — Resilience Suite

Governing Statement:: Structure under stress.

Visitor Prompt:: See how living systems absorb shocks and maintain integrity.

QR Entry:: https://life.aquarium.com/suite/18

LIFE Principle:: Resilience is not the absence of damage. It is the capacity to absorb damage and continue functioning. Every ecosystem, every organism, every organization has a resilience threshold.

Suite 19 — Extreme Climate Suite

Governing Statement:: Specialization for survival.

Visitor Prompt:: Marvel at organisms engineered by evolution for the harshest conditions.

QR Entry:: https://life.aquarium.com/suite/19

LIFE Principle:: Deserts, deep ocean, polar ice, volcanic vents — life exists in all of them. The machinery required to survive extremes reveals what biological engineering is capable of when the constraint is absolute.

Suite 20 — LIFE Synthesis Suite

Governing Statement:: All systems connect.

Visitor Prompt:: Bring together every principle explored across the LIFE experience.

QR Entry:: https://life.aquarium.com/suite/20

LIFE Principle:: Perception, signal, constraint, energy, adaptation, resilience — none of these exist in isolation. This suite connects every principle into a unified view of how living systems work.

Suite Deployment Rules

1. No suite is themed by staff. Suite themes are set by the owner and locked. Staff may not "theme" or redecorate any suite.
2. No content added without owner authorization. Every element in a narration suite — physical, digital, visual — is owner-approved.
3. QR codes must be functional. If a QR code in any suite is broken or leads to the wrong destination, report it immediately. Do not create a replacement.
4. Damage is reported immediately. Any damage to suite elements — signs, displays, props, screens — is reported. Do not attempt repair without authorization.
5. Suite progression is intentional. The 20 suites follow a designed sequence from Perception to Synthesis. The order is not arbitrary and must not be rearranged.

Thematic Progression Map

Phase	Suites	Theme
Perception & Signal	1–5	How do we detect and interpret reality?
Language & Systems	6–8	How do we organize and cooperate?
Pattern & Constraint	9–10	What rules govern behavior?
Resource & Energy	11–12	How is energy allocated?
Adaptation & Perspective	13–15	How does environment shape identity?
Transformation & Limits	16–17	When does specialization become fragility?
Resilience & Extremes	18–19	What survives stress and why?
Synthesis	20	How does it all connect?

PART 21

SIGNAGE SYSTEM: REPLACEMENT MAP

This section defines the complete signage replacement protocol for both facilities. It is an active execution plan, not a reference document. Work proceeds according to the execution order specified at the end of this section.

Global Removal Rule

All non-canonical signs are removed.

A canonical sign is any sign that was:

1. Designed within the approved signage system
2. Produced and installed under owner authorization
3. Listed in an approved sign inventory

Any sign that does not meet all three criteria is non-canonical. Non-canonical signs are removed immediately and replaced according to this map. There are no exceptions for signs that are "mostly right" or "still accurate." If it was not produced under this system, it comes down.

Signage Categories — 5 Types

Category 1 — Species / Encounter Signs

Format:: Vertical totem format

Content:: Species-specific information structured around the 8-question totem framework (see Part 19).

Design:

- Vertical orientation
- Species common name at top — large, readable at 3 feet

- Scientific name in italics below common name
- 8 micro-questions with answers
- Reflection question at bottom
- Facility mark (logo or wordmark) at base

Placement:: At or adjacent to the primary viewing position for each species exhibit.

Authority:: Content approved by owner. Design produced by approved designer. Installation by authorized staff only.

Category 2 — Instructional Signs

Format:: Icon-based vinyl

Content:: Step-by-step instructions for guest interactions — feeding, touching, behavior near exhibits.

Design:

- Icon-first — the visual instruction is the primary communication
- Minimal text — confirm what the icon shows, do not write paragraphs
- High contrast — readable in all facility lighting conditions
- Consistent icon family across all instructional signs

Placement:: At the point of interaction — beside or directly above the action zone.

Authority:: Content approved by operations manager and owner. Design produced by approved designer.

Category 3 — Educational / Conservation Signs

Format:: Grouped by zone

Content:: Conservation context, ecological relationships, and systems-level information relevant to the species or habitat zone.

Design:

- Zone-grouped — multiple signs that work together in a zone, not standalone panels
- Mechanism-first language — describe systems, not sentiments
- No persuasion, no moral imposition

- Consistent with LIFE education framework (Part 7)

Placement:: Within the themed zone, not crowded at a single point. Visitors encounter them progressively as they move through the space.

Authority:: Content approved by education lead and owner. Design produced by approved designer.

Category 4 — Commercial / Pricing Signs

Format:: Standardized vinyl

Content:: Encounter costs, feeding fees, and any purchase-required experiences.

Design:

- Standardized format — consistent layout and typography across all pricing signs
- Clear hierarchy: experience name → price → what is included
- No decorative excess — pricing signs are functional
- Updated immediately when pricing changes. No outdated pricing signs remain in place.

Placement:: At the point of decision — where a guest would decide whether to purchase.

Authority:: Content approved by owner. No pricing sign goes up without owner sign-off.

Category 5 — Directional / Wayfinding Signs

Format:: Icon + arrow

Content:: Location labels, directional arrows, zone identifiers.

Design:

- Icon communicates the destination type (restroom, exit, exhibit area, food, etc.)
- Arrow indicates direction — cardinal or relative (left, right, forward)
- Text label confirms the icon (in cases where icon alone is insufficient)
- Consistent icon family with instructional signs

Placement:: At all decision points — any location where a guest must choose a direction.

Authority:: Layout approved by operations manager and owner.

TV Screen Reset — 3 Roles

Every TV screen in the facility is assigned one of three roles. A screen may not serve multiple roles simultaneously. A screen without an assigned role is off or dark until assigned.

Role 1 — Instruction

Purpose:: Tells guests what to do at an interactive exhibit or station.

Content:: How-to steps, behavioral guidelines, safety reminders for specific interactions.

Format:: Looping, short, icon-supported. Not a lecture. Gets the guest to the right behavior in under 30 seconds.

Example placements:: Beside touch pools, at feeding stations, at entry to interaction zones.

Role 2 — Education

Purpose:: Delivers LIFE system content — species information, ecological relationships, system explanations.

Content:: Mechanism-first. Observation-based. Aligned with the 8-question totem framework where applicable.

Format:: Looping content that a guest can engage for 30 seconds or 3 minutes — depth available if they stay, not required if they don't.

Example placements:: Near large exhibit windows, in narrative suite zones, in waiting areas adjacent to exhibits.

Role 3 — Ambient

Purpose:: Creates atmosphere and environmental immersion without competing for attention.

Content:: Nature footage, slow-paced visuals related to the zone's theme, sound-optional (designed to be effective with or without sound).

Format:: Looping ambient video. No text. No calls to action. No pricing or promotions.

Example placements:: Hallways, transitional spaces, high-traffic areas where guests are moving, not stopping.

Execution Order — 6 Steps

Work proceeds in this order. Steps are not skipped. Steps are not reversed.

Step 1 — Inventory all current signs.

Document every sign in the facility — type, location, content, condition. Photograph each one. This is the baseline removal list.

Step 2 — Flag all non-canonical signs for removal.

Using the global removal rule, identify every sign that was not produced under the approved system. Tag it for removal.

Step 3 — Remove all non-canonical signs.

Pull all tagged signs. Do not replace with handwritten or temporary alternatives. A blank wall is preferable to a wrong sign.

Step 4 — Install canonical signs by category.

Begin with Category 1 (Species/Encounter) — the highest-priority visitor-facing content. Then Category 2 (Instructional), then Category 3 (Educational), then Category 4 (Commercial/Pricing), then Category 5 (Directional/Wayfinding).

Step 5 — Reset all TV screens.

Assign each screen its role. Load approved content. Confirm looping and display are functioning. Document each screen's assigned role and content.

Step 6 — Audit and verify.

Walk the entire facility. Confirm every sign is canonical, correctly placed, and undamaged. Confirm every screen is displaying its assigned content. Report any gaps.

Paused Items

The following signage items are identified but not yet in active production or installation. They are paused pending owner review, design completion, or content finalization:

- Narration Suite signage (content locked in Part 20 — physical production pending)
- QR code-integrated totem panels (digital link confirmation required)

- Outdoor wayfinding system (layout finalization pending)
- Spanish-language sign variants (translation review required)

Paused items are not ignored — they are tracked and will enter the execution order when unpaused by the owner.

PART 22

ROLE-BASED OPERATIONAL PACKETS

These packets are the role-specific summaries of operational requirements. Every role has a packet. Every staff member reads their packet in addition to the universal sections (Parts 1–5, Part 12).

These packets do not replace the full document. They are structured summaries for daily reference.

MANAGER PACKET V1

Role:: Manager

Version:: V1

Authority:: Managers operate within systems designed and approved by the owner. Managers execute, verify, and escalate. Managers do not redesign systems.

What Managers Own

1. Diet Photo Review — Daily

Every diet photo submitted by staff must be reviewed every day. The review is not passive. You look at the photo, confirm it meets the standards (full plate, correct items, top-down, captioned), and respond: "Verified – [Your Initials]."

If the photo is incorrect:

- Do not ignore it.
- Do not silently approve it.
- Require correction immediately.
- Document the error.

No silent approvals. Silence is not verification.

2. Feeding Baseline Enforcement

No animal in a guest-interaction exhibit is fed to fullness before guest hours. You enforce this. You confirm it. If you observe pre-loading, you correct it immediately.

3. Exhibit Integrity Checks

Walk the exhibits. Confirm:

- Nothing has been added to any exhibit without authorization.
- No animals have been relocated without authorization.
- No personal items, decorations, or unauthorized objects are present.
- Exhibits are clean, properly lit, and animals appear healthy.

If anything is out of standard, correct it or report it immediately. Do not wait.

4. Escalation Responsibility

When a staff member error occurs:

- Strike 1: Correct it. Document it internally.
- Strike 2: Issue the formal warning. Conduct the one-on-one. Supervise the retraining. Collect written acknowledgment.
- Strike 3: Initiate disciplinary review. Consult the owner.

You do not skip strikes. You do not handle Strike 3 without the owner.

5. Photo Verification of Cleaning

Food prep areas must be photographed clean after each use. You verify this the same way you verify feeding photos — actively, not passively.

6. Zero Waste Enforcement

Nothing goes in the trash. If you observe food waste going in the trash, correct it immediately. Assign it to the Zero Waste Food Loop (Part 6). Document the deviation.

7. Field Trip Oversight

Field trip groups follow LIFE protocol (Part 7). Staff do not make up facts. Staff do not create their own materials. Education runs through the system — not through improvised presentations. If a field trip deviates from protocol, correct it.

What Managers Cannot Do

- Approve diet changes
- Approve exhibit modifications

- Approve software adoption
- Approve vendor contracts or purchases above threshold
- Sign any agreement on behalf of the facility
- Modify any system without owner authorization

When in doubt: STOP → ASK → WAIT.

Manager Daily Checklist (Summary)

- [ ] All diet photos reviewed and responded to
- [ ] No unauthorized pre-loading confirmed
- [ ] Exhibit integrity walk completed
- [ ] Cleaning photos verified
- [ ] Zero Waste compliance confirmed
- [ ] Any errors documented and escalated per the 3-Strike Ladder

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EMPLOYEE PACKET V1

Role:: Animal Care / Floor Staff / Education Staff

Version:: V1

Authority:: Staff execute tasks as assigned, within written procedures, with photo verification at each step.

Your Core Obligations

1. Follow the Written Diet. Photograph Every Feeding.

The diet sheet tells you what to prepare. You prepare exactly that — nothing more, nothing less. No substitutions. No guessing. No "close enough."

After preparation and before delivery:

- Take a clear photo — top-down, all items visible, proper lighting.
- Caption: Animal Name – Date – Your Initials.
- Post the photo to the designated channel.

Then deliver the food.

If something is wrong with the diet sheet — not your job to fix it. Your job: stop, report it to a manager.

2. Diet Language Rules (Ammon Covino — Owner Directive)

- No grams or metric measurements. Use observable quantities: "one mango," "a handful of pellets," "one plantain."
- No "mixed greens." Use specific names: mulberry leaves, approved local browse.
- No romaine lettuce. Ever.
- Describe what you are doing in observable terms. Not abstract categories.

3. Do Not Feed to Fullness Before Guest Hours (Guest-Interaction Exhibits)

If the exhibit you are responsible for involves guest feeding or guest interaction, you do not pre-load the animal before guests arrive. The guest feeding is part of the animal's daily intake. You plan accordingly. See Part 17 for your specific exhibit's addendum.

4. Photo Verification Is Mandatory

No photo = not done. This applies to:

- Every feeding
- Every food prep area after cleaning
- Any task the system requires documentation for

If you completed a task but forgot to photograph it, you re-do the documentation step. You do not skip it.

5. Zero Waste

Nothing goes in the trash. Uneaten food goes to the correct bin — Primary Compost, Controlled, or Reject. If you are unsure where something goes — STOP → ASK. Do not guess. Do not throw it away.

6. Exhibit Integrity

- Do not add anything to any exhibit.
- Do not remove anything from any exhibit without authorization.
- Do not relocate any animal.
- If something looks wrong in an exhibit — report it. Do not fix it yourself.

7. Cleanliness

- Acrylic windows: blue rags + distilled water ONLY.
- Virkon for exhibit disinfection.
- Food prep areas: cleaned, sanitized, photographed after every use.

- Water dishes: carbon filtered water where specified.

If you do not know the correct cleaning method — ASK before you do anything.

8. Field Trip Rules

If you are working with a field trip group:

- Do not make up facts about animals.
- Direct questions you cannot answer to the QR code or a manager.
- Do not create your own educational materials.
- Follow the LIFE framework — observation-first, no persuasion, no ideology.

9. Escalation

If you see something wrong — a sick animal, a safety issue, a protocol violation — you report it. You do not wait. You do not fix it yourself if it is outside your authorization. You report it to your manager immediately.

If you are not sure about anything: STOP → ASK → WAIT.

Employee Daily Reference (Summary)

- Prepare feeding per the written diet sheet. No substitutions.
- Photograph every feeding before delivery.
- Do not pre-load guest-interaction animals before guest hours.
- Send all uneaten food to the Zero Waste loop — not the trash.
- Keep exhibits exactly as they are — do not add, remove, or modify anything.
- Use correct cleaning products for every surface.
- Photograph food prep areas after cleaning.
- Report anything wrong — immediately.

STAFF PACKET V1

Role:: General Staff (all roles, including part-time, seasonal, new hires)

Version:: V1

Authority:: General staff execute assigned tasks within written procedures. No task is valid without a claim.

The Basics — What Every Staff Member Must Know

1. Nothing Without Authorization

If you were not told to do it, if it is not written down, if nobody approved it — you do not do it. This is Rule 1 of the entire system (Part 1). It is not negotiable.

2. Claim Your Tasks

Tasks are posted. You claim the ones assigned to you. When you claim a task, you own it. If nobody claims a task, that is an emergency — escalate immediately.

3. Photo Everything the System Requires

When a task requires photo documentation — feeding, cleaning, food prep — you take the photo. You caption it correctly. You post it. You do not skip this step.

4. Baseline Feeding Rules

For any exhibit you are assigned to:

- Use the written diet sheet. Exactly.
- No substitutions. Ever.
- No feeding to fullness before guest hours at guest-interaction exhibits.
- Photo before delivery.

5. Photo Verification

Every feeding = every photo. This is how we know it was done correctly. No photo = not done.

6. Exhibit Integrity

Do not touch, move, add to, or remove anything from any exhibit. If it looks wrong, report it.

7. Cleanliness

Do not guess at cleaning products. Ask if unsure. Blue rags + distilled water for acrylic. Virkon for exhibits. Photograph food prep surfaces after cleaning.

8. Field Trip Rules

If guests or students ask you a question you cannot answer — say so. Do not make up information. Use this script: "That is a great question. Scan the QR code for more information, or I can find someone who can help."

9. Escalation Rules

You see something wrong:

1. Do not fix it yourself unless you are authorized.

2. Report it to your manager immediately.
3. If your manager is unavailable and it is urgent — go up the chain. Do not wait.

10. When in Doubt

STOP → ASK → WAIT.

This is not a weakness. This is the system. Guessing is how animals get hurt and systems break.

Staff Quick Reference Card

Situation	What You Do
Not sure if you should do something	STOP → ASK → WAIT
No diet sheet available	Do not feed. Report to manager.
Diet item unavailable	Do not substitute. Report to manager.
Animal looks wrong or unwell	Report to manager immediately. Do not wait.
Something added to an exhibit	Report to manager. Do not touch it.
Food prep area dirty	Clean it. Photograph it after cleaning.
Uneaten food	Zero Waste loop. Not the trash.
Guest asks a question you cannot answer	QR code or find a manager. Do not guess.
Guest complains	Acknowledge. Do not argue. Escalate to manager.

PART 23

HOUSTON OPERATIONS SOP: CLARITY & AUTHORITY

This SOP governs all operational decision-making at the Houston Interactive Aquarium & Animal Preserve. It defines authority, communication, and enforcement structures. It is not a guideline. It is operational law for this location.

If something contradicts this SOP, this SOP wins — unless the owner has issued a more recent written directive.

SOP Rule 1: The Owner's Voice Is Final

All operational decisions ultimately rest with the owner, Ammon Covino. When the owner speaks — in person, in writing, or through a documented directive — that is the decision. It does not require debate. It does not require consensus. It requires execution.

If a manager, a senior employee, or any other individual at this facility overrides, ignores, or delays acting on an owner directive without explicit owner permission to do so, that is a system violation. It is treated as such — immediately.

Owner directive received = owner directive executed.

SOP Rule 2: Management Authority Is Delegated, Not Autonomous

Managers at this facility operate within authority that has been delegated by the owner. They do not hold independent operational authority.

What this means:

- Managers direct daily task execution within existing approved procedures.
- Managers enforce existing rules.
- Managers do not create new rules.
- Managers do not modify systems.
- Managers do not approve purchases, contracts, or vendor relationships.
- Managers do not override diet sheets.

- Managers do not approve exhibit modifications.

If a manager believes a rule needs to change or a system needs adjustment, the manager brings it to the owner. The owner decides. Until the owner decides, the existing system stands.

SOP Rule 3: Staff Authority Is Execution Only

Staff execute assigned tasks within written procedures. That is the entire scope of staff authority.

Staff do not:

- Approve anything
- Create procedures
- Modify exhibits
- Change diets
- Make purchasing decisions
- Adopt software or tools
- Sign anything on behalf of the facility
- Make promises to vendors, guests, or partners

If a staff member is asked to do any of the above by anyone — including a manager — the answer is: "I need to check with the owner first."

This is not insubordination. This is the system.

SOP Rule 4: No Verbal Approvals Without Written Follow-Up

Verbal instructions and verbal approvals exist in this facility as operational communication. They are valid for immediate action direction. They are not valid as permanent records.

Any verbal instruction or approval that affects:

- Diet or feeding
- Purchasing
- Exhibit modification
- Vendor relationships
- Staff discipline

...must be followed by written documentation within 24 hours. If it is not documented, it did not happen as a formal system change.

"He told me to do it" is not sufficient if "he" did not document it.

SOP Rule 5: Photo Documentation Is the Proof Standard

In all matters where documentation is required, the proof standard at this facility is photographic verification. Written logs are secondary. Verbal reports are tertiary. Photos are primary.

This applies to:

- All feedings
- All food prep area cleanings
- All exhibit anomalies (photograph before touching anything)
- All delivery and receiving (photograph all incoming supplies)
- All signage changes (photograph before and after)

If you are uncertain whether a situation requires a photo — photograph it. The cost of an unnecessary photo is zero. The cost of a missing photo can be significant.

SOP Rule 6: Escalation Is Mandatory, Not Optional

When a situation exceeds the authority or capability of the person facing it, escalation is mandatory. It is not a suggestion. It is not optional if you "think you can handle it."

The escalation path is:

1. Staff → Manager
2. Manager → Owner

Situations that require immediate owner escalation (do not wait for manager intermediate step if manager is unreachable):

- Any animal health emergency
- Any guest safety incident
- Any situation involving potential legal liability
- Any media inquiry

- Any request to sign a contract or agreement
- Any situation where the system has broken down in a way that cannot be corrected at the manager level

When in doubt about whether to escalate — escalate.

SOP Rule 7: System Changes Require Written Owner Approval

Nothing about this facility's operating system changes without written owner approval. This includes:

- Diet modifications
- Exhibit changes
- Staff scheduling structures
- Vendor relationships
- Software and technology adoption
- Pricing changes
- Education content changes
- Signage changes
- Partnership terms

If a change is needed urgently, you contact the owner and wait for written approval. "We needed to do it quickly" is not an acceptable reason for making an unauthorized system change. If speed is required, escalate faster — do not bypass the approval process.

SOP Rule 8: Communication Standards

All operational communication at this facility adheres to these standards:

Clarity:: State what you are communicating, what action is required, and by when. Do not make people decode your message.

Channel integrity:: Use the designated communication channel for each type of communication. Do not deliver operational decisions through unofficial channels and then expect them to be treated as official.

No gossip, no politics:: Communication about operational matters stays operational. Staff disagreements about procedures go through proper escalation channels. They do not become

social conflicts played out in the facility.

Written before binding:: Anything that binds the facility to a course of action — a commitment, an agreement, a directive that changes how we operate — is written before it is binding.

Response expectations:: If you receive a communication that requires a response, respond. Silence is not acceptance. Silence is ambiguity. Ambiguity is a system failure.

SOP Rule 9: Animal Welfare Is the Non-Negotiable Baseline

Every operational decision at this facility — scheduling, staffing, purchasing, event planning, guest programming — is evaluated against one baseline: does this protect animal welfare?

If a decision compromises animal welfare, it does not happen. There are no exceptions:

- No event scheduling that disrupts animal care operations
- No staffing reductions that leave animal care understaffed
- No purchasing decisions that compromise diet quality
- No guest programming that prioritizes entertainment over animal health

This is not sentiment. This is the operating logic of the facility. The animals are the product. The product must be protected above everything else.

Animal welfare is not a department. It is the operating standard for every department.

SOP Rule 10: Consistency Is Accountability

The same rule applied consistently to everyone is what makes a system fair. Rules applied inconsistently create instability, resentment, and system failure.

At this facility:

- The 3-Strike Escalation Ladder applies to staff AND managers.
- Photo verification is required from everyone — not just from new hires.
- The authorization rule applies to everyone — not just to people whose authority is being questioned.
- Diet protocols are followed by everyone who handles food — not just by whoever was last corrected.

Consistency is not rigidity. Consistency is accountability made visible.

A rule that only applies to some people is not a rule. It is a suggestion. We do not operate on suggestions.

PART 16

Employee Sign-Off

I have read this entire document and I understand:

- ✓ The Owner's Top Ten — Operational Standards
- ✓ The Owner's Top Ten — Guest-Facing Standards
- ✓ How tasks work (claim-based system)
- ✓ The authorization model
- ✓ The diet verification protocol
- ✓ The Zero Waste Food Loop
- ✓ The cleaning standards
- ✓ The education system (LIFE)
- ✓ The exhibit integrity rules
- ✓ The technology governance rules
- ✓ The purchasing authority rules
- ✓ The workspace standards
- ✓ The Do Not List
- ✓ The escalation process: correction, warning, disciplinary action

I had the opportunity to ask questions. I understand that if I am unsure about something, I will stop and ask before acting.

Employee Name (print): _____

Employee Signature: _____

Date: _____

Conducted By (print): _____

Conductor Signature: _____

Date: _____

This document is retained by management. A copy may be provided to the employee upon request.

LIFE + Studio Giraffe Integrated Operational System

Owner: Ammon Covino | San Antonio Aquarium | Houston Interactive Aquarium & Animal Preserve | Effective: April 2026